

Empirically Measuring Privacy Over the NIST CRC Deidentified Data Archive

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Knexus Research

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NIST

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National Institute of Standards and Technology

This talk focuses on a NIST metrology program for data **deidentification techniques**.

Deidentification includes any processing to microdata that produces microdata in the same schema and is *intended* to be resistant to individual reidentification: SDC, synthetic data, differential privacy.

Since February 2023 we've been running a massive community benchmarking and meta-analysis project, collecting metrics, algorithms and data samples from stakeholders, researchers and statistical agencies around the world— and making them all freely available and easy to use.

In this talk we'll be presenting our recent and upcoming work on **empirical privacy metrics**.



NIST's Collaborative Research Cycle: Benchmarking Deidentification

Deidentification Techniques Evaluated in this Talk:

Differentially Private Synthetic Data

- SmartNoise MST ($E = 1, 10$)
- SmartNoise AIM ($E = 1, 10$)

Non-DP Statistical Model Synthetic Data

- R Synthpop CART

Proprietary AI (non-DP) Synthetic Data

- AINDO
- MostlyAI
- Anonos

Statistical Disclosure Control

- SDCMicro Cell Suppression (k-6, small vs large quasi ID set)
- 40% Subsample

Sanity Checks: Complete ground truth data, complete withheld data

NIST's Collaborative Research Cycle: Benchmarking Deidentification

Deidentification Techniques Evaluated in this Talk:

All synthetic data approaches shown here have reasonable utility on this data set

Library	Algorithm	Utility: Subsample Equivalence
aindo-synth	aindo-synth	30.0
Anonos Data Embassy SDK	Anonos Data Embassy SDK	20.0
MostlyAI SD	MostlyAI SD	30.0
rsynthpop	cart	40.0
smartnoise-synth	aim	20.0
smartnoise-synth	aim	60.0
smartnoise-synth	mst	10.0
smartnoise-synth	mst	10.0
sdcmicro	kanonymity all-features	1
sdcmicro	kanonymity-quasiID	20
subsample	subsample_40pcnt	40.0

Data Record



Features A



Features B



The pieces of our problem

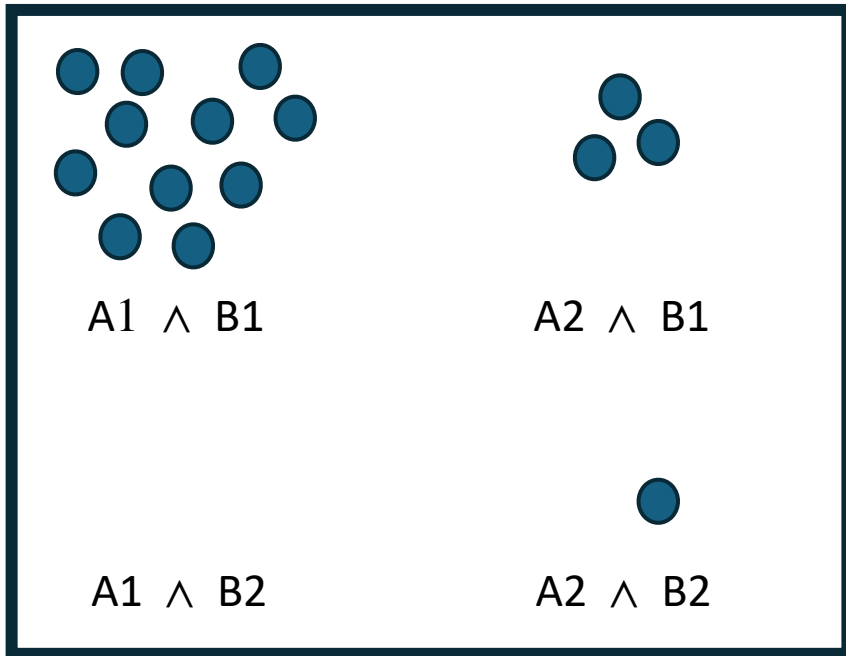
Features A



Features B



Input Real Data



The pieces of our problem

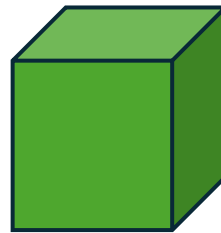
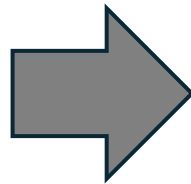
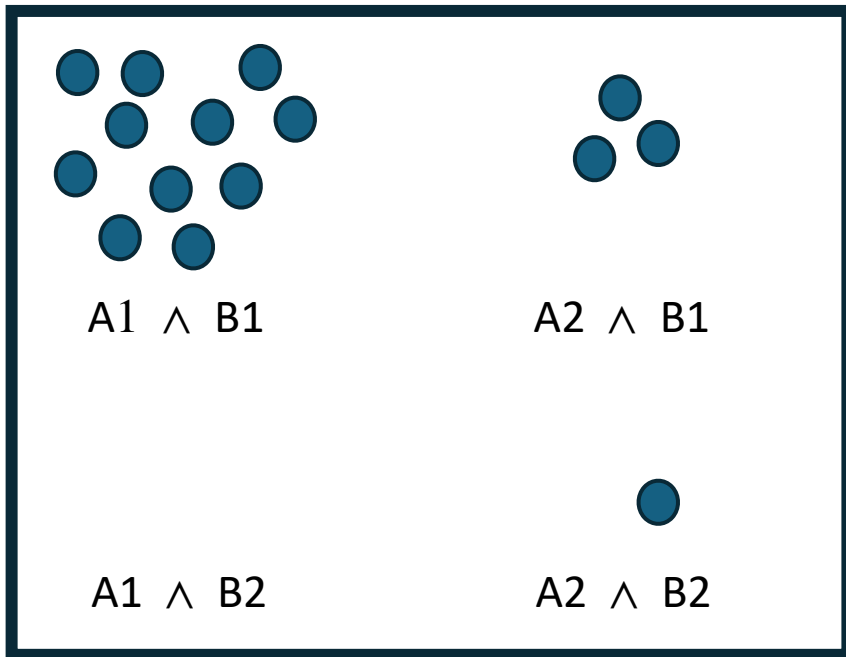
Features A



Features B



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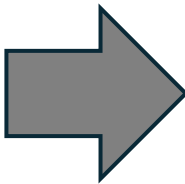
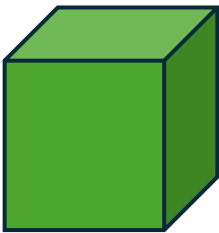
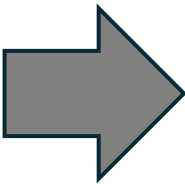
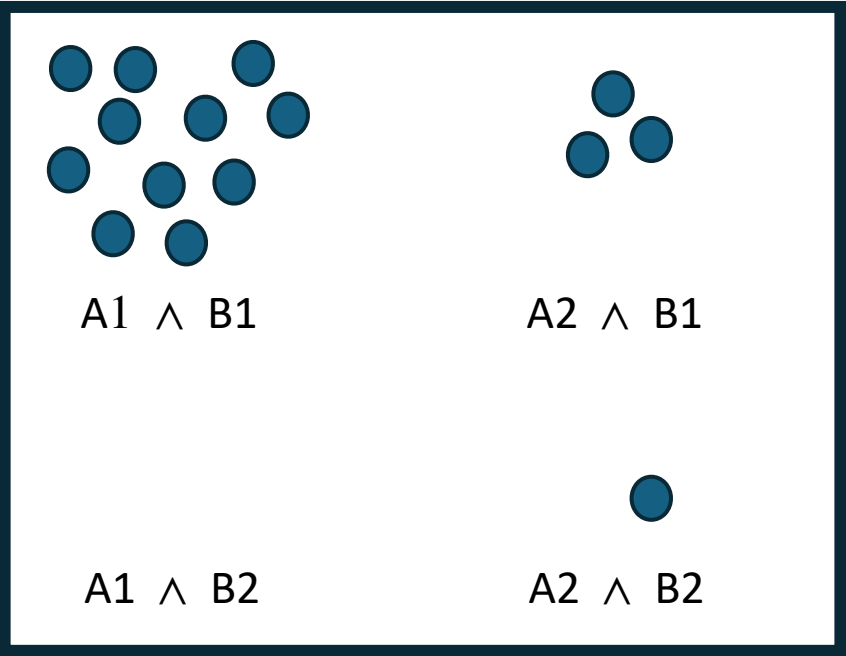
Features A



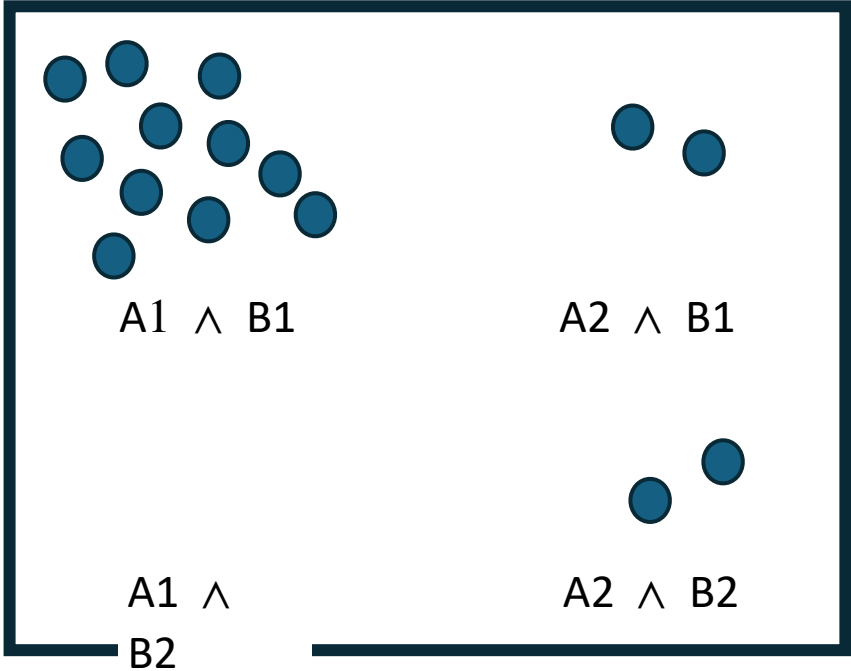
Features B



Input Real Data



Deidentified Data



The pieces of our problem

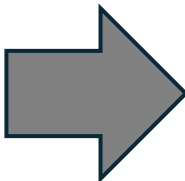
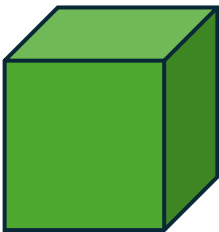
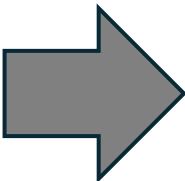
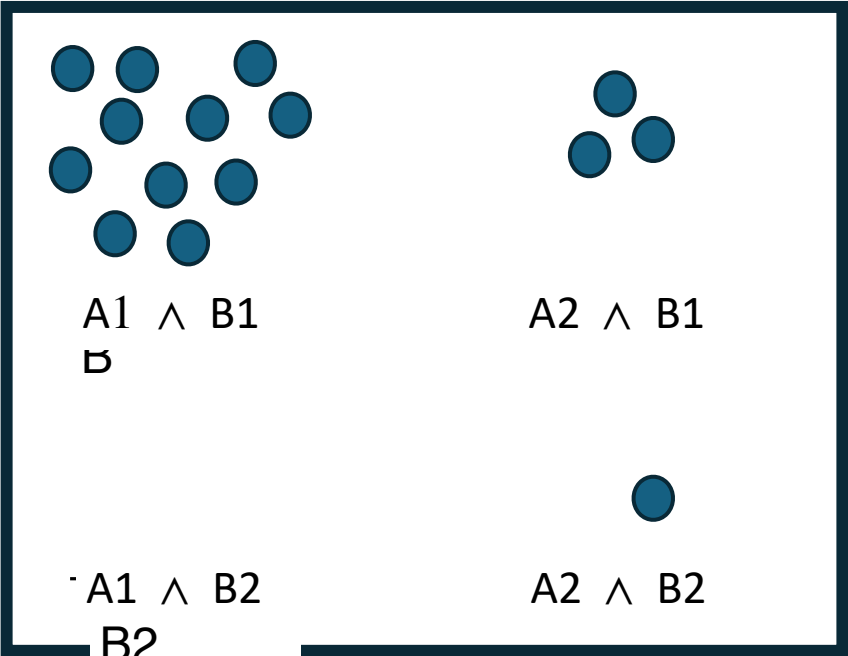
Features A



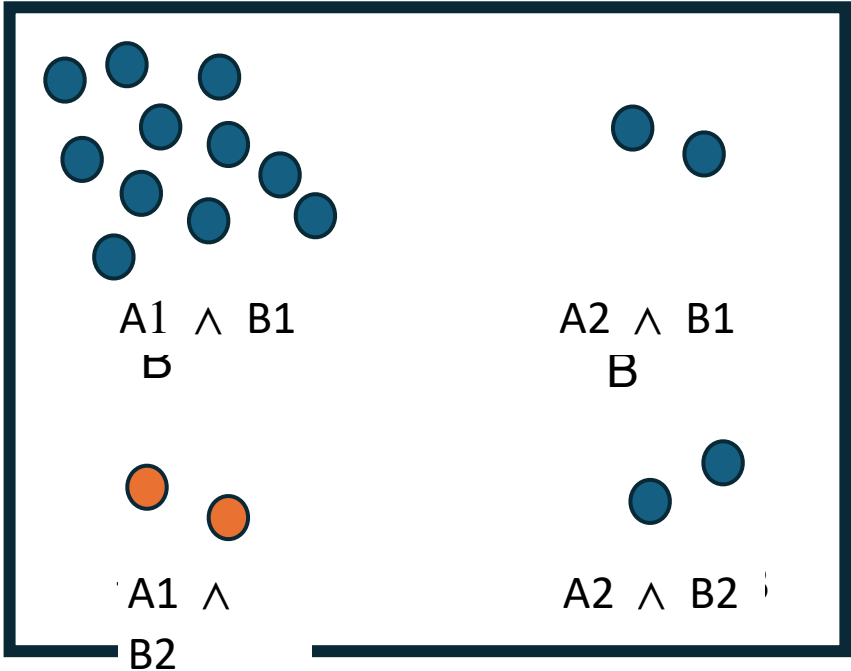
Features B



Input Real Data



Deidentified Data



The pieces of our problem

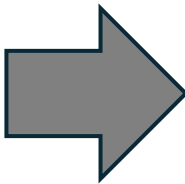
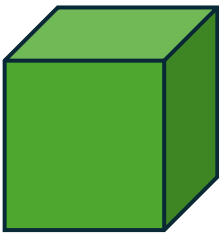
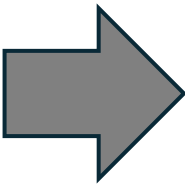
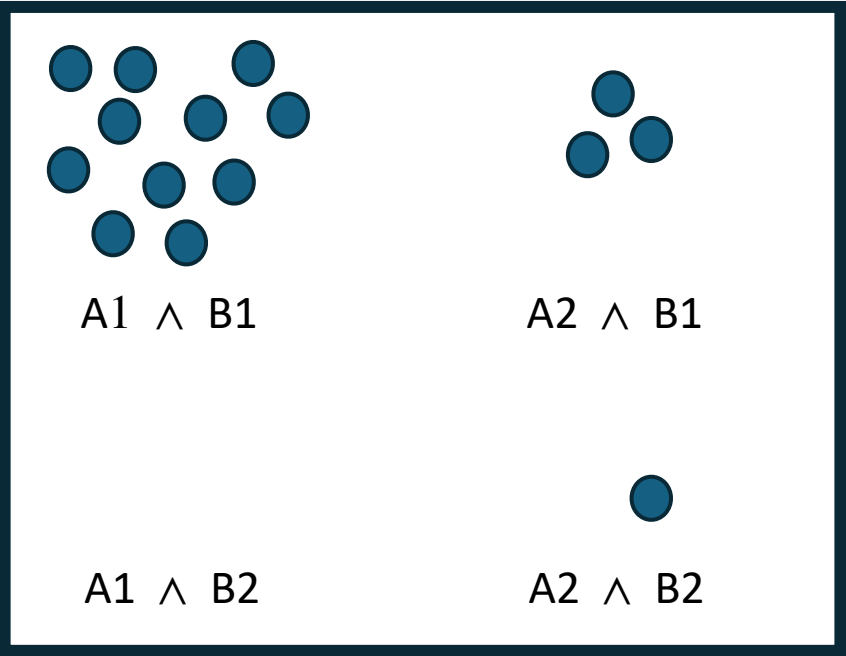
Features A



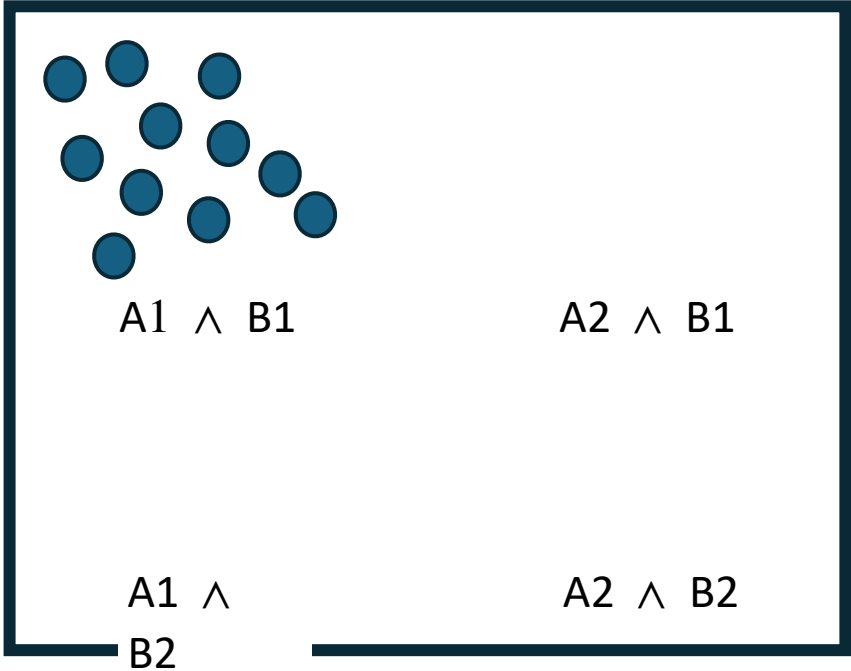
Features B



Input Real Data



Deidentified Data



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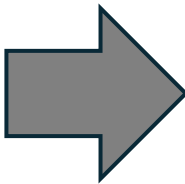
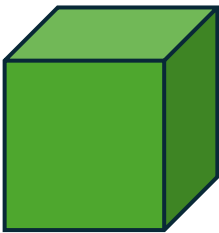
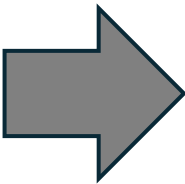
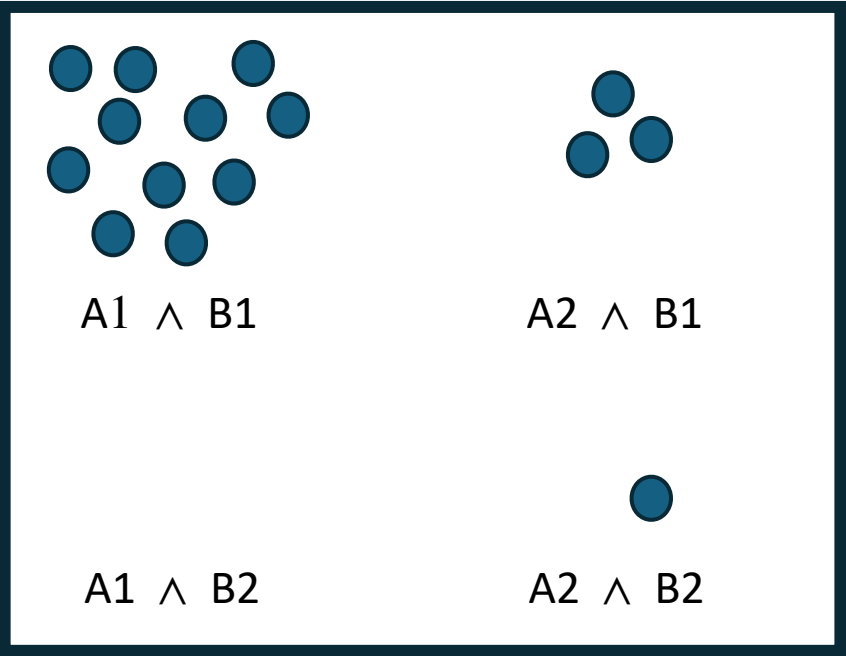
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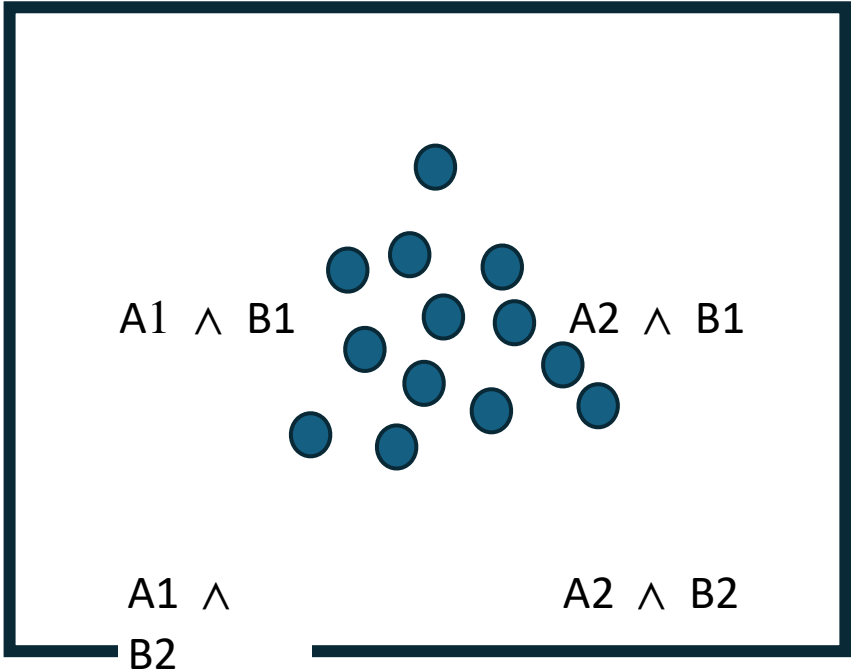
Features B



Input Real Data



Deidentified Data



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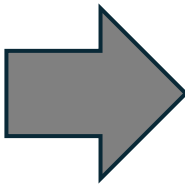
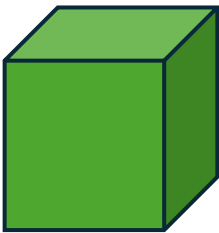
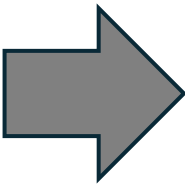
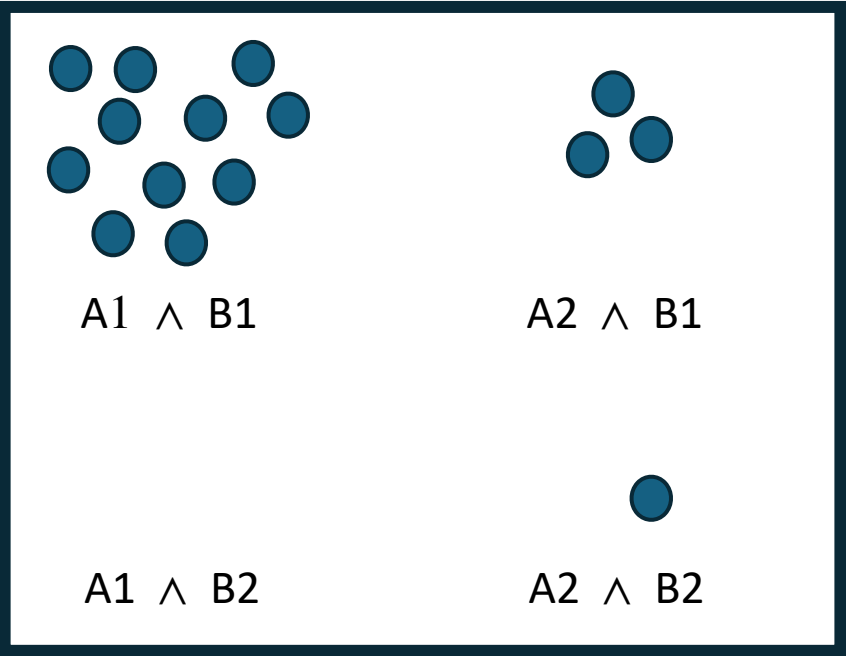
Features A



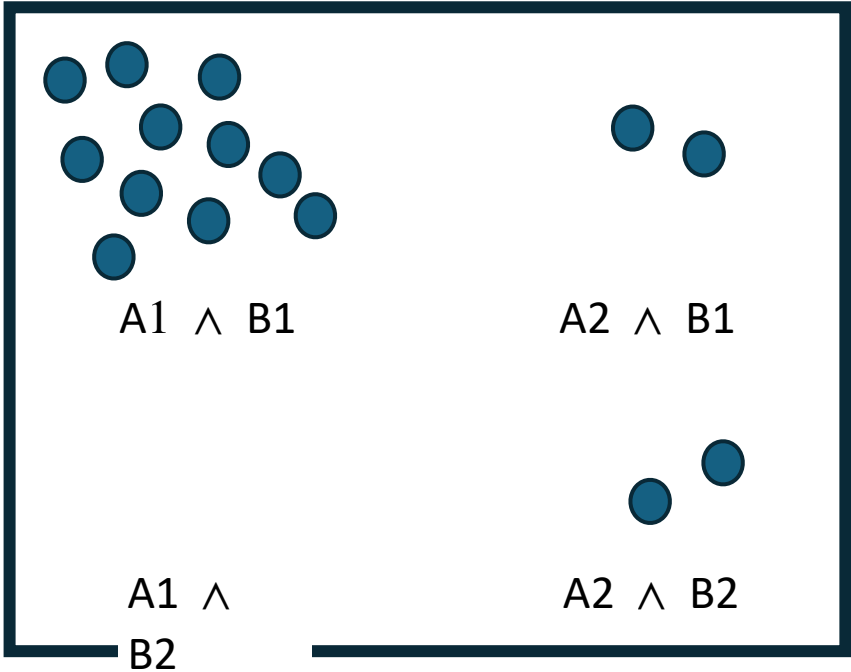
Features B



Input Real Data

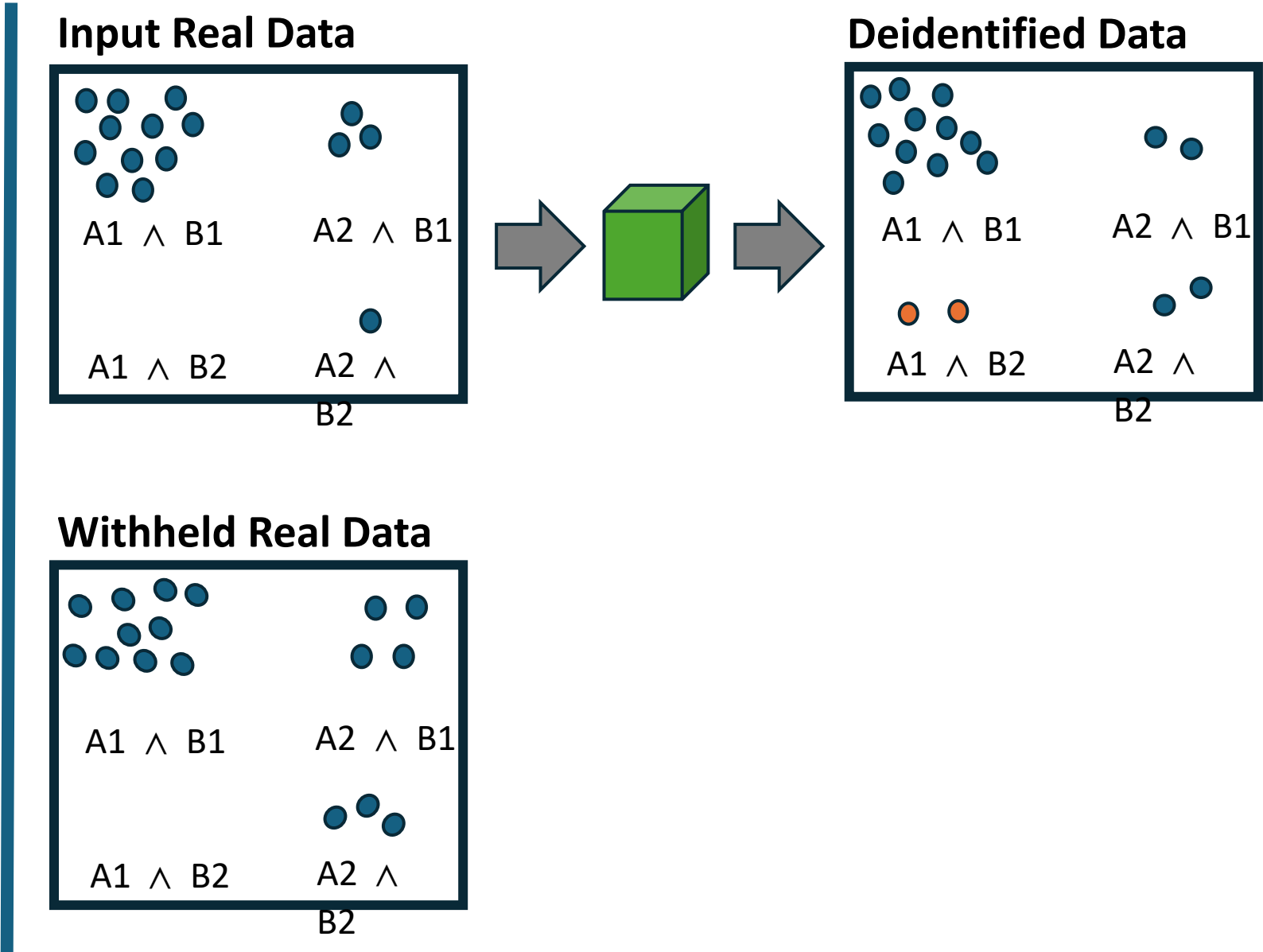


Deidentified Data



The pieces of our problem

Measuring Privacy for Deidentified Data:

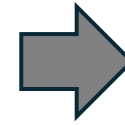
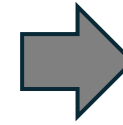
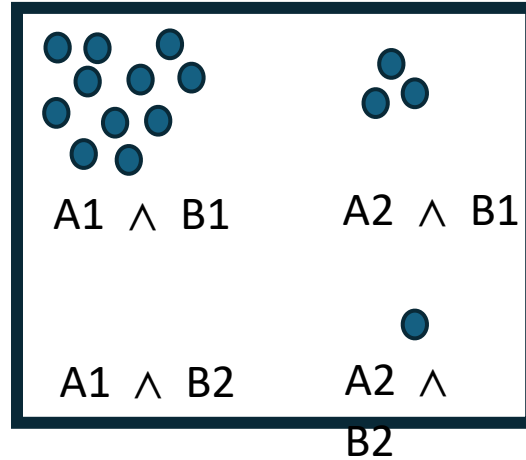


The pieces of our problem

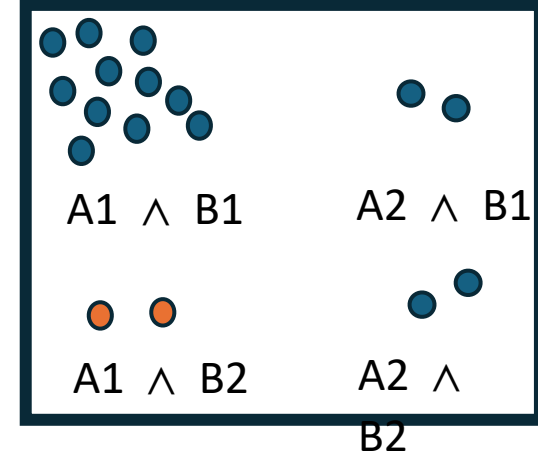
Measuring Privacy for Deidentified Data:

Could anyone use the deidentified data to do something bad that would make us regret having released it?

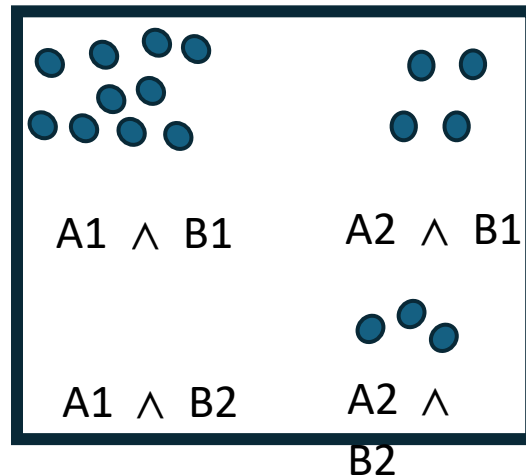
Input Real Data



Deidentified Data



Withheld Real Data



Can I use the released data to learn **something invasive** about you?

Attribute Privacy:

If I know someone's **identifying features A**, could I use the deidentified data to infer invasive features B

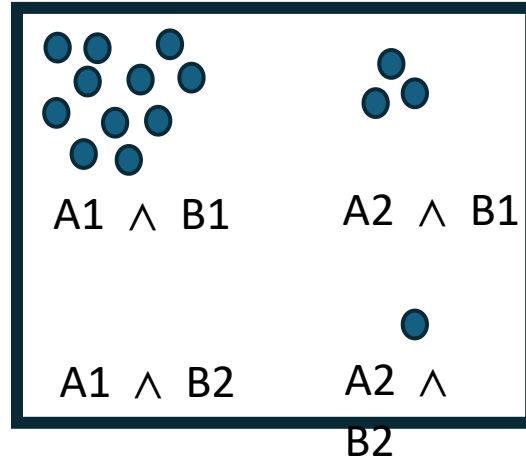
(1) Better than I could using the withheld data?

[Anonymeter: Inference]

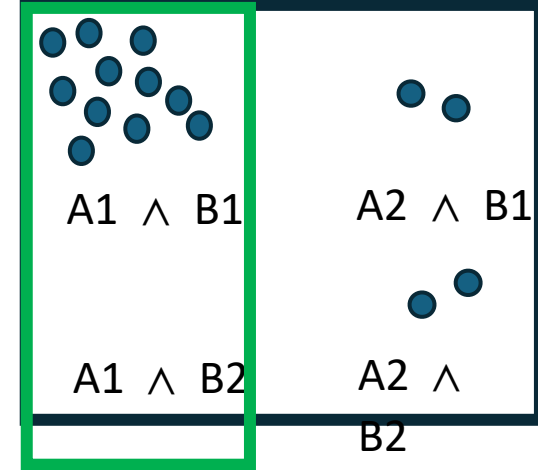
(2) Worse than I could using the input real data?

[Synthpop: DiSCO]

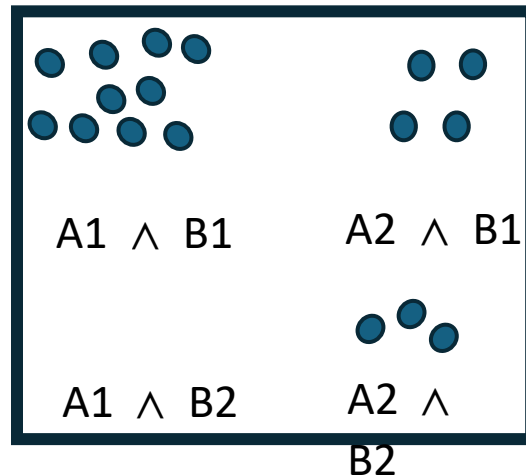
Input Real Data



Deidentified Data



Withheld Real Data



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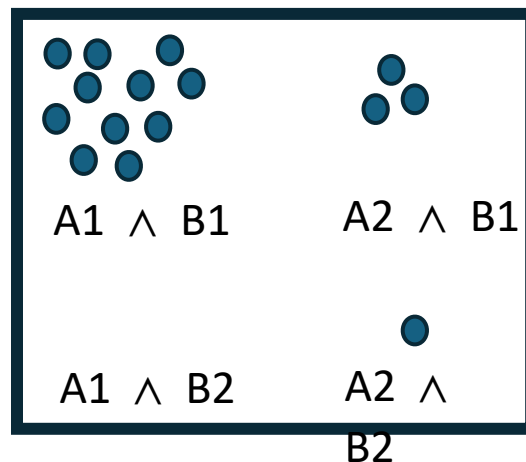
(1) better than I could using the withheld data?

[Anonymeter: Inference]

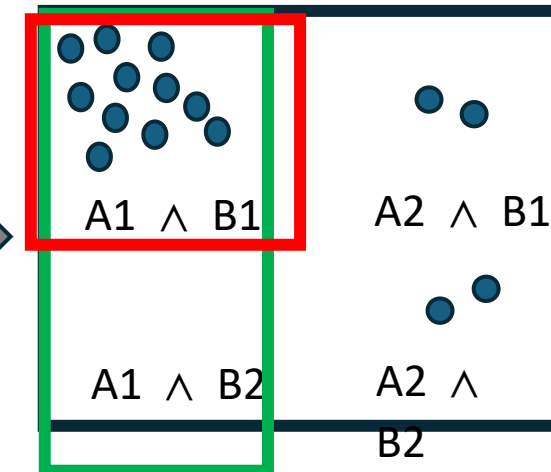
(2) worse than I could using the input real data?

[Synthpop: DiSCO]

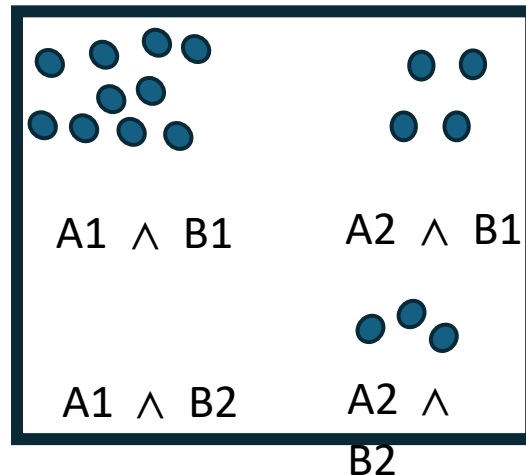
Input Real Data



Deidentified Data



Withheld Real Data



Can I use the released data to learn **something invasive** about you?

Attribute Privacy:

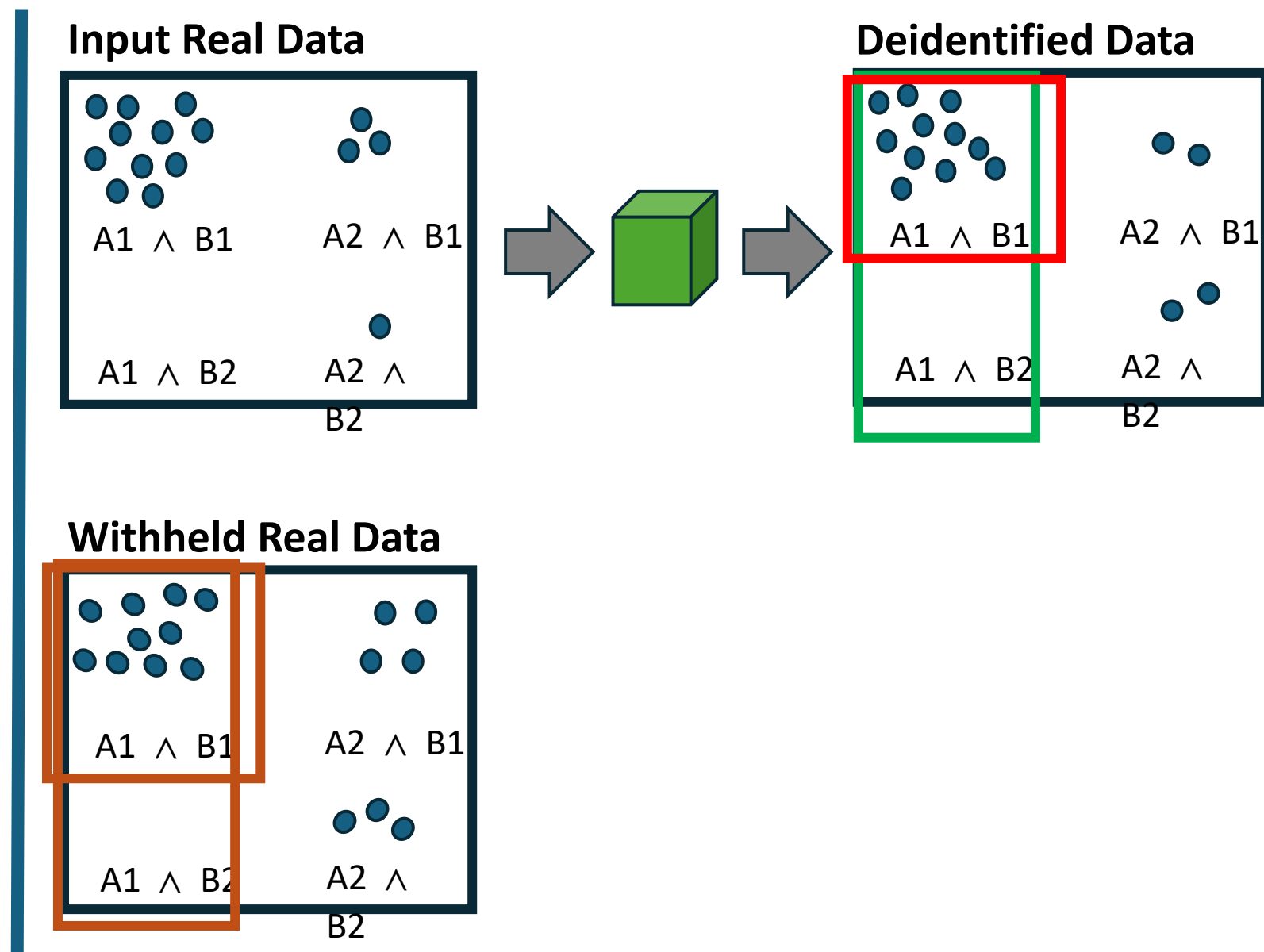
If I know someone's **identifying features A**, could I use the deidentified data to infer **invasive features B**

(1) better than I could using the **withheld data**?

[Anonymeter: Inference]

(2) worse than I could using the input real data?

[Synthpop: DiSCO]



Can I use the released data to learn **something invasive** about you?

Attribute Privacy:

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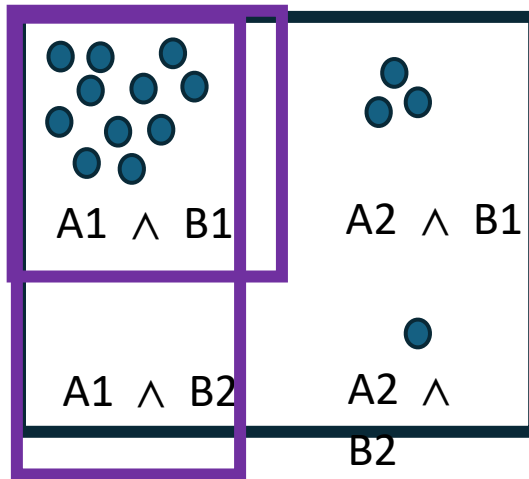
(1) better than I could using the withheld data?

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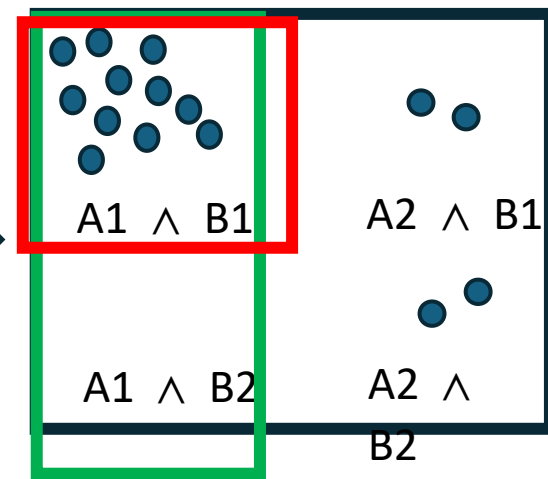
(2) worse than I could using the **input real data**?

[Synthpop: DiSCO]

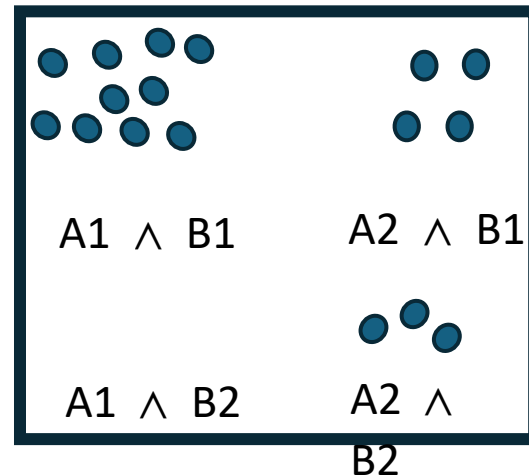
Input Real Data



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Withheld Real Data



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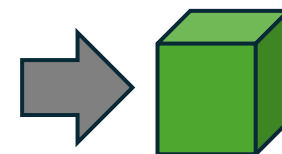
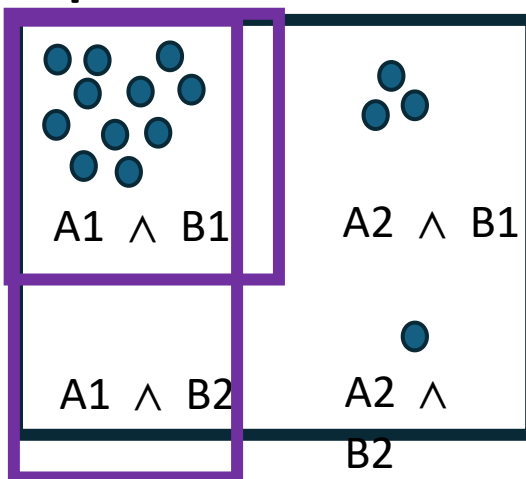
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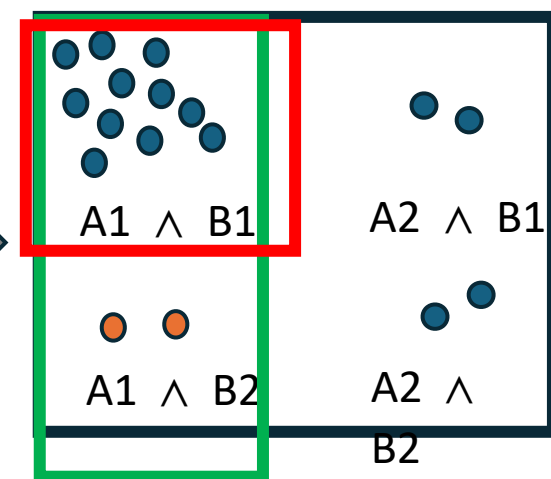
(2) worse than I could using the **input real data**?

[Synthpop: DiSCO]

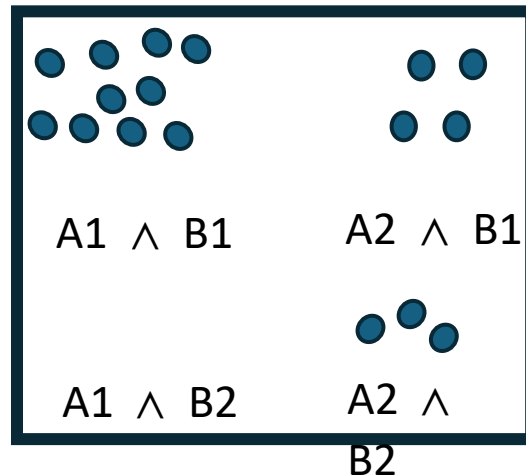
Input Real Data



Deidentified Data



Withheld Real Data

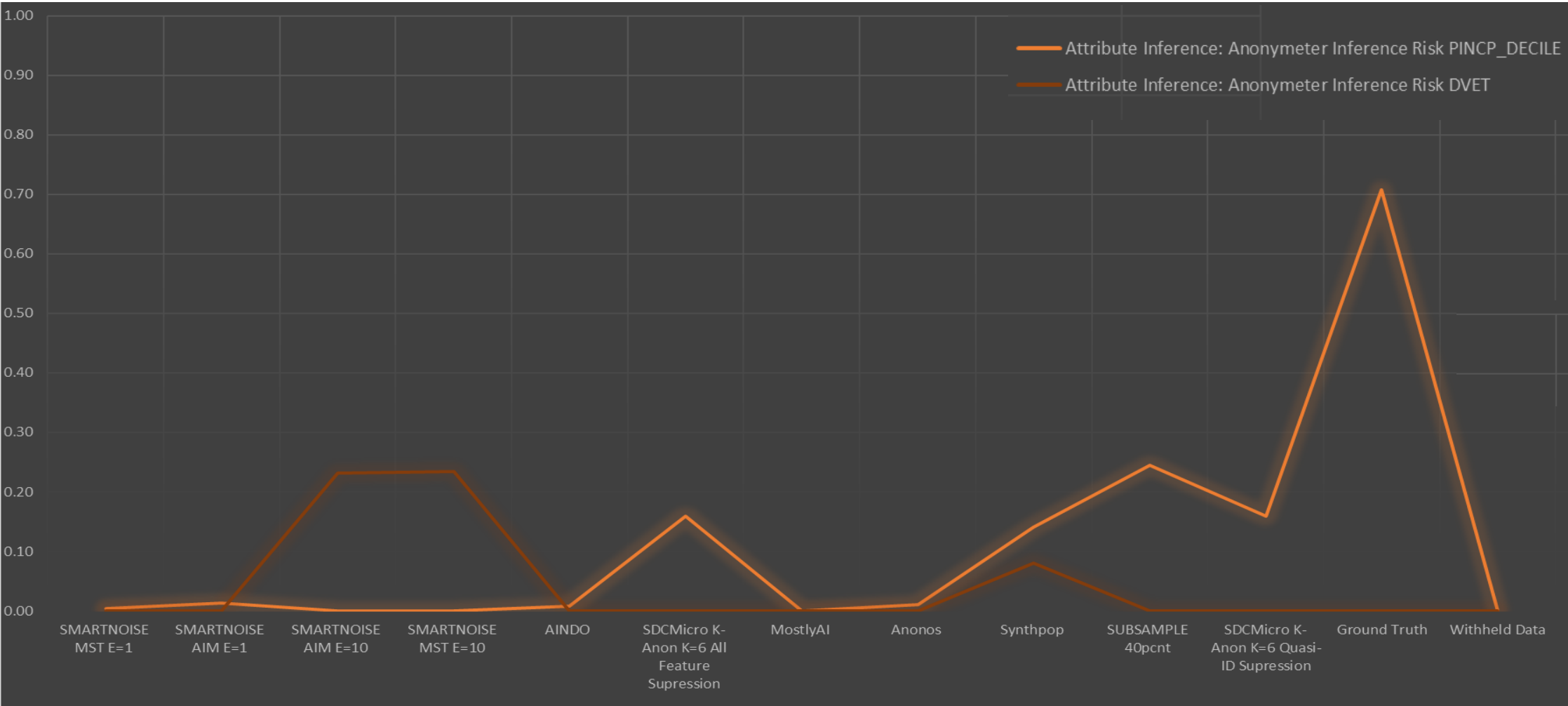


Can I use the released data to learn **something invasive** about you?

Attribute Privacy: If I know someone’s identifying features A, could I use the deidentified data to infer invasive features B

(1) **better than I could using the withheld data? [Anonymeter: Inference]**

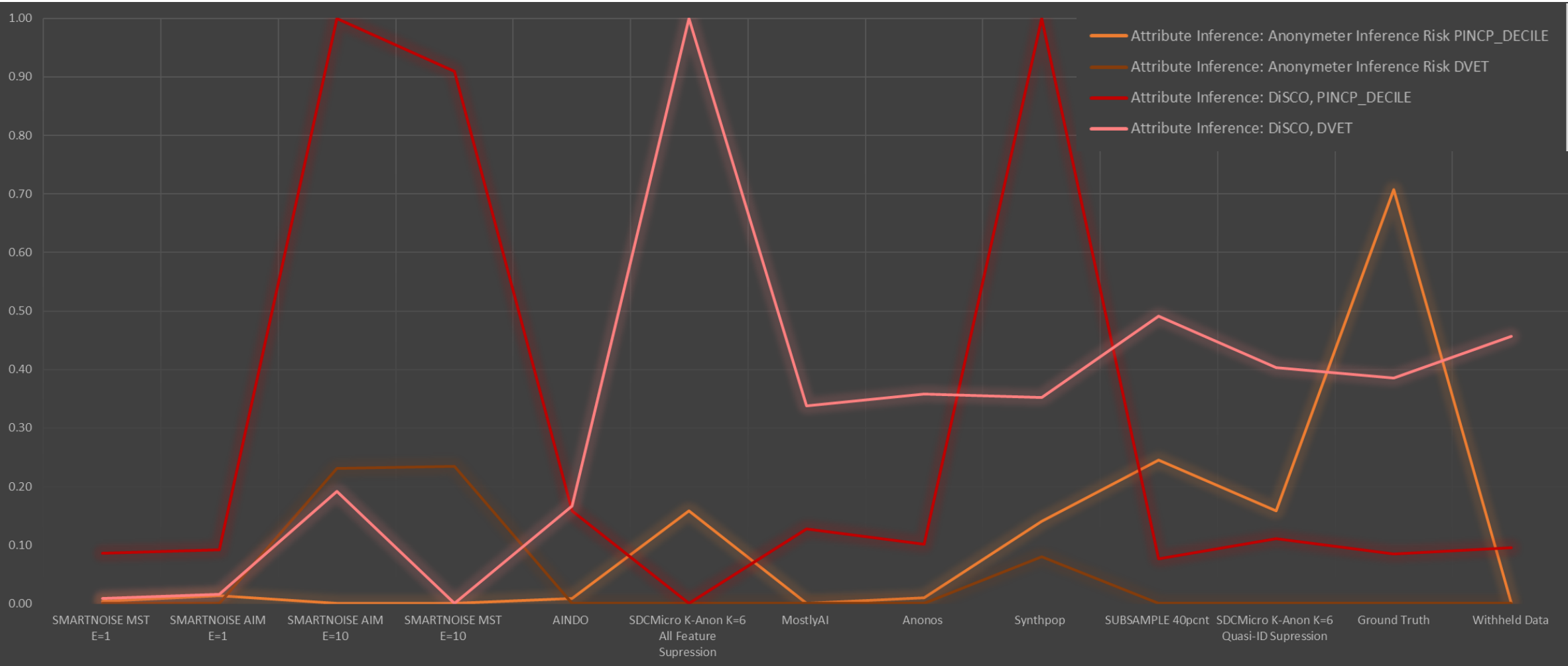
(2) better than I could using the input real data? [Synthpop: DiSCO]



Can I use the released data to learn **something invasive** about you?

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- (1) **better than I could using the withheld data?** [Anonymeter: Inference]
- (2) **better than I could using the input real data?** [Synthpop: DiSCO]



Can I use the released data to learn whether **you were in the input data**?

Membership Privacy:

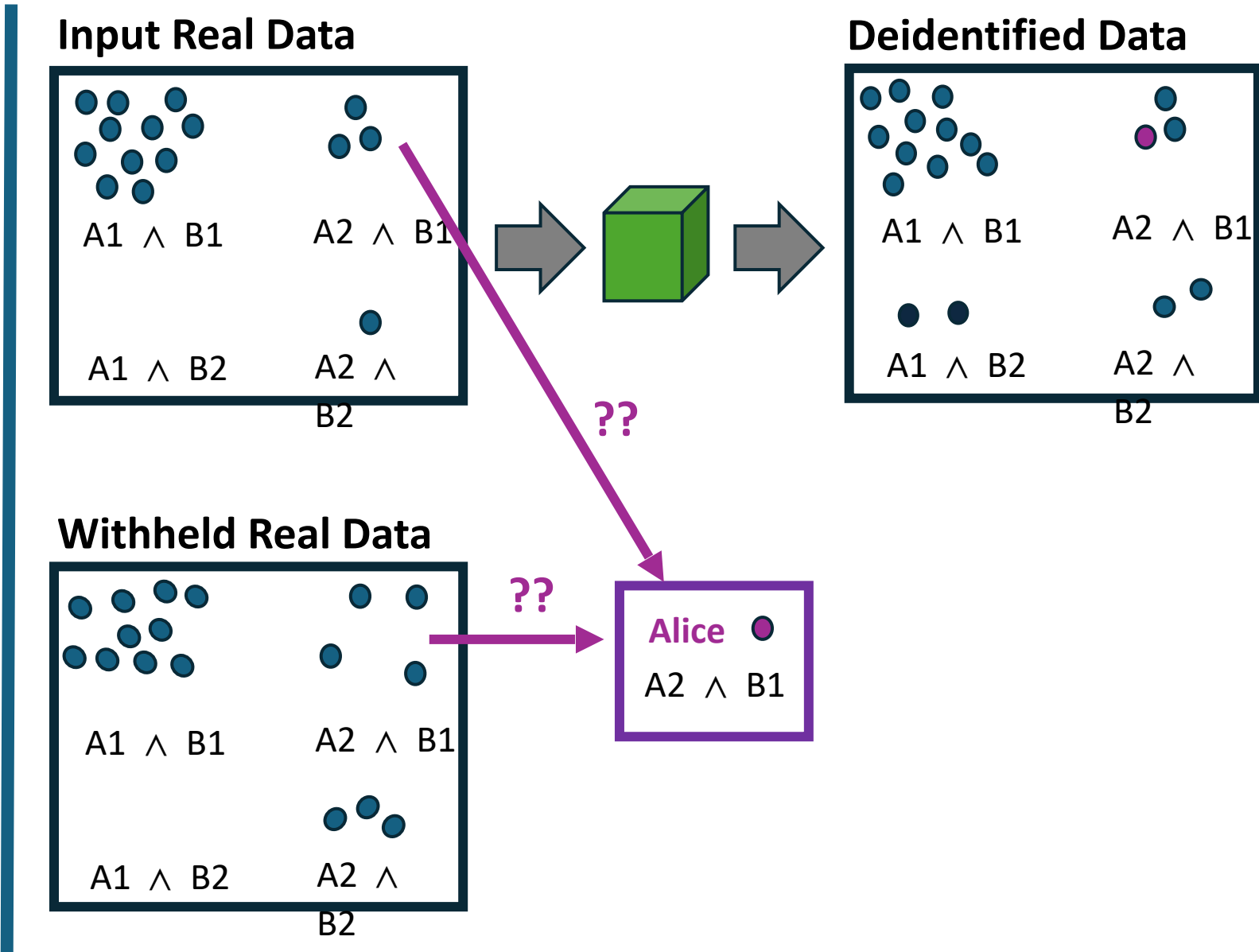
If I know someone's record, can I tell if they were in the (problematic) input data set?

(1) by checking if vulnerable outlier records look very similar to deidentified records?

[Tumult: Empirical Leakage]

(2) by checking if deidentified data has dense spots where the withheld data doesn't

[SynthCity: DoMIAS Prior]



Can I use the released data to learn whether **you were in the input data**?

Membership Privacy:

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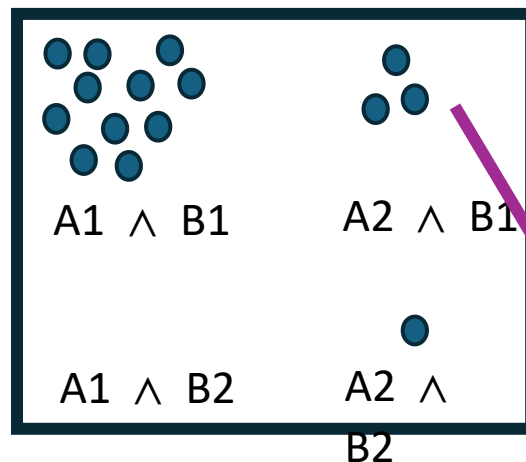
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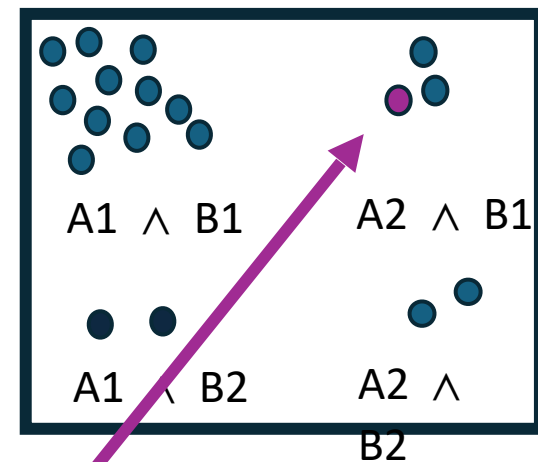
(2) by checking if deidentified data **has dense spots** where the withheld data doesn't

[SynthCity: DoMIAS Prior]

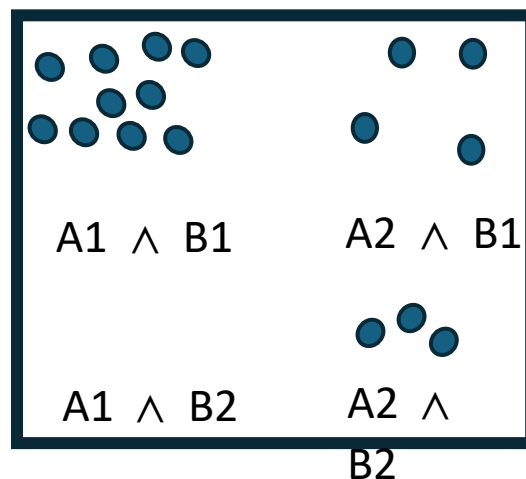
Input Real Data



Deidentified Data



Withheld Real Data



??

!!

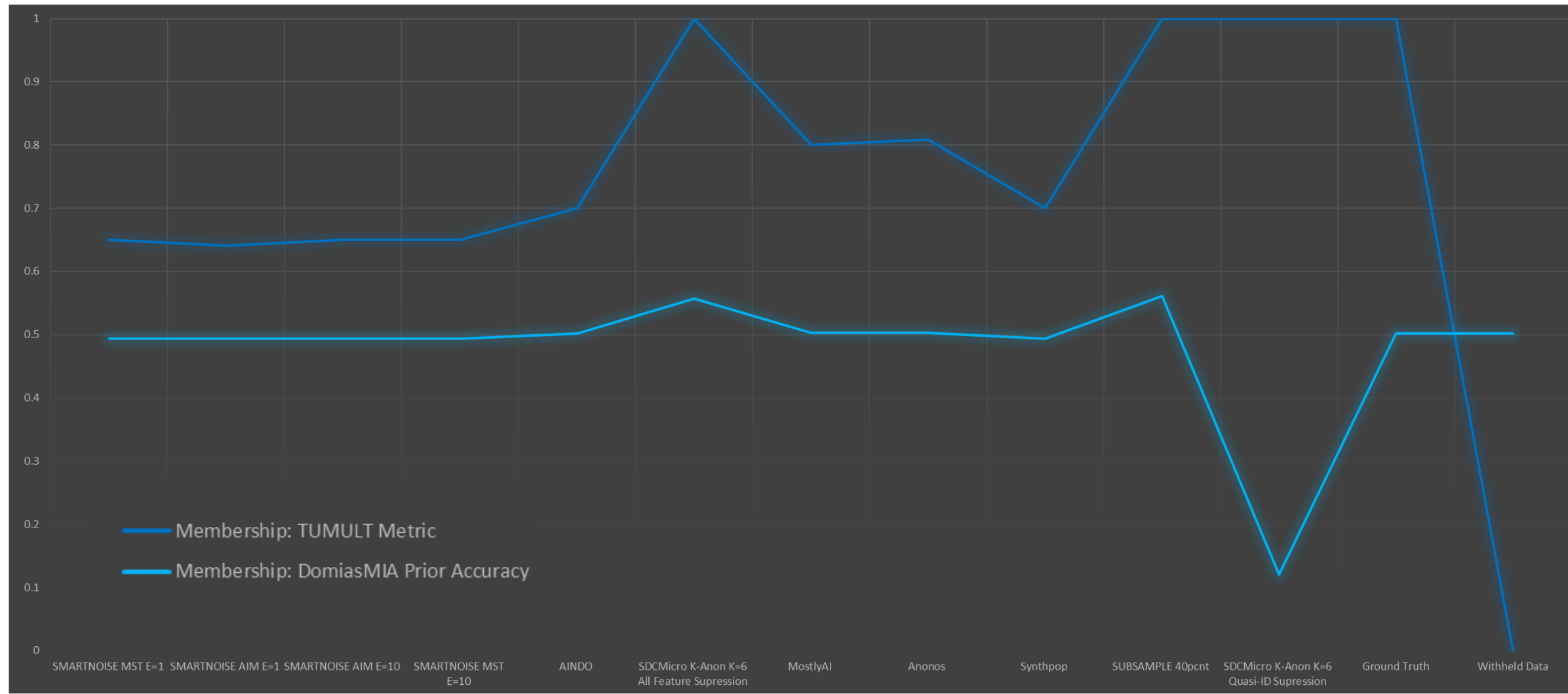
Alice ●
A2 $\wedge$ B1

Can I use the released data to learn whether **you were in the input data?**

Membership Privacy: If I know someone's record, can I tell that they were in the input data set

(1) by checking if vulnerable outlier records look similar to deidentified records? [Empirical Leakage]

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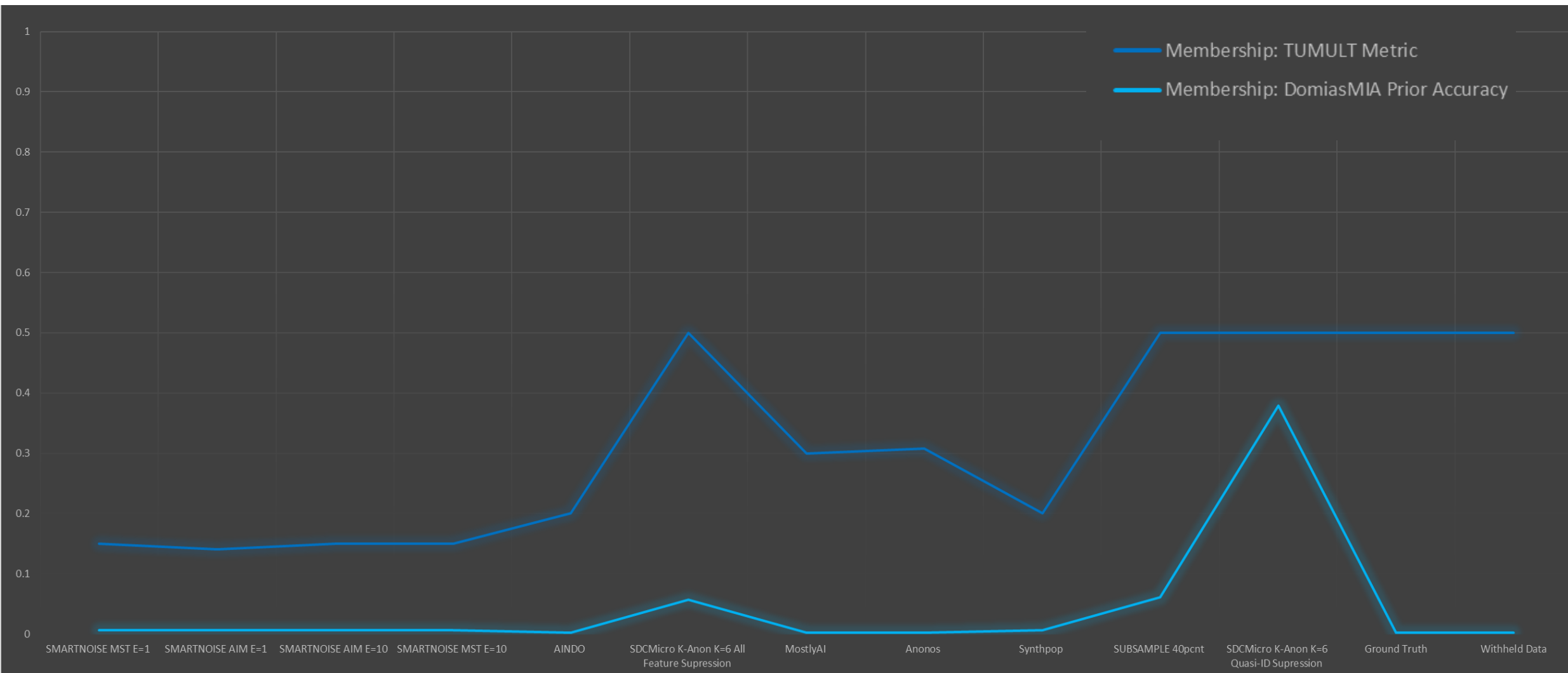


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(1) by checking if vulnerable outlier records look similar to deidentified records? [Empirical Leakage]

(2) by checking if deidentified data has dense spots where the withheld data doesn't [DoMIAS Prior]



Can I use the released data to.... *find you*?

Reidentification Risk:

An anonymization concept:
Can the deidentified data help
me find your record because...

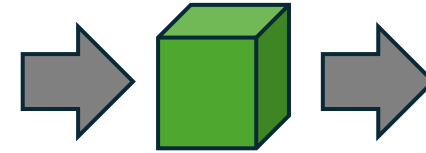
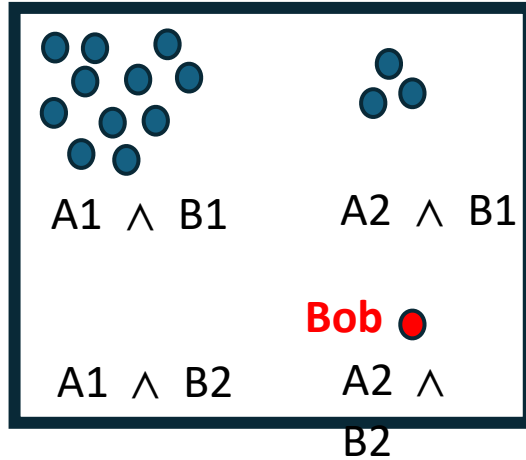
(1) your unique outlier record
survived deidentification?

[CRC: Unique Exact Match]

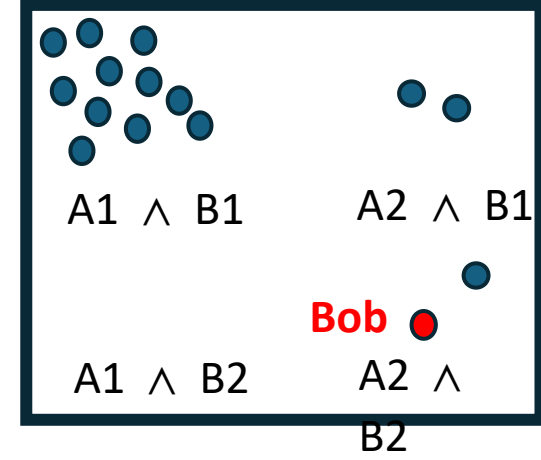
(2) the released deidentified data
forms a bridge between two real
data sets with features A and
features B?

[Anonymeter: Linkability]

Input Real Data



Deidentified Data



Can I use the released data to.... *find you*?

Reidentification Risk:

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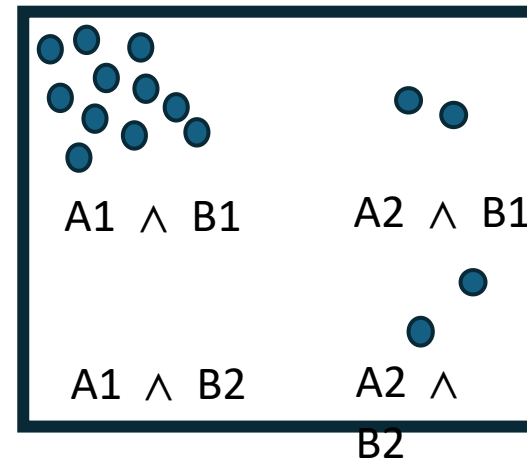
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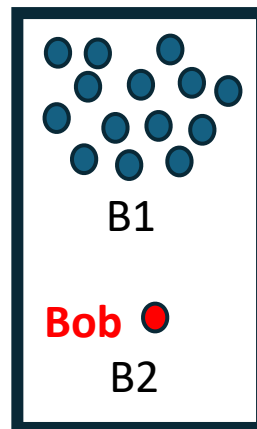
(2) the released deidentified data
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[Anonymizer: Linkability]

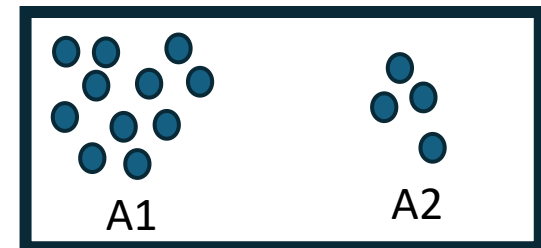
Deidentified Data



Real Data B



Real Data A



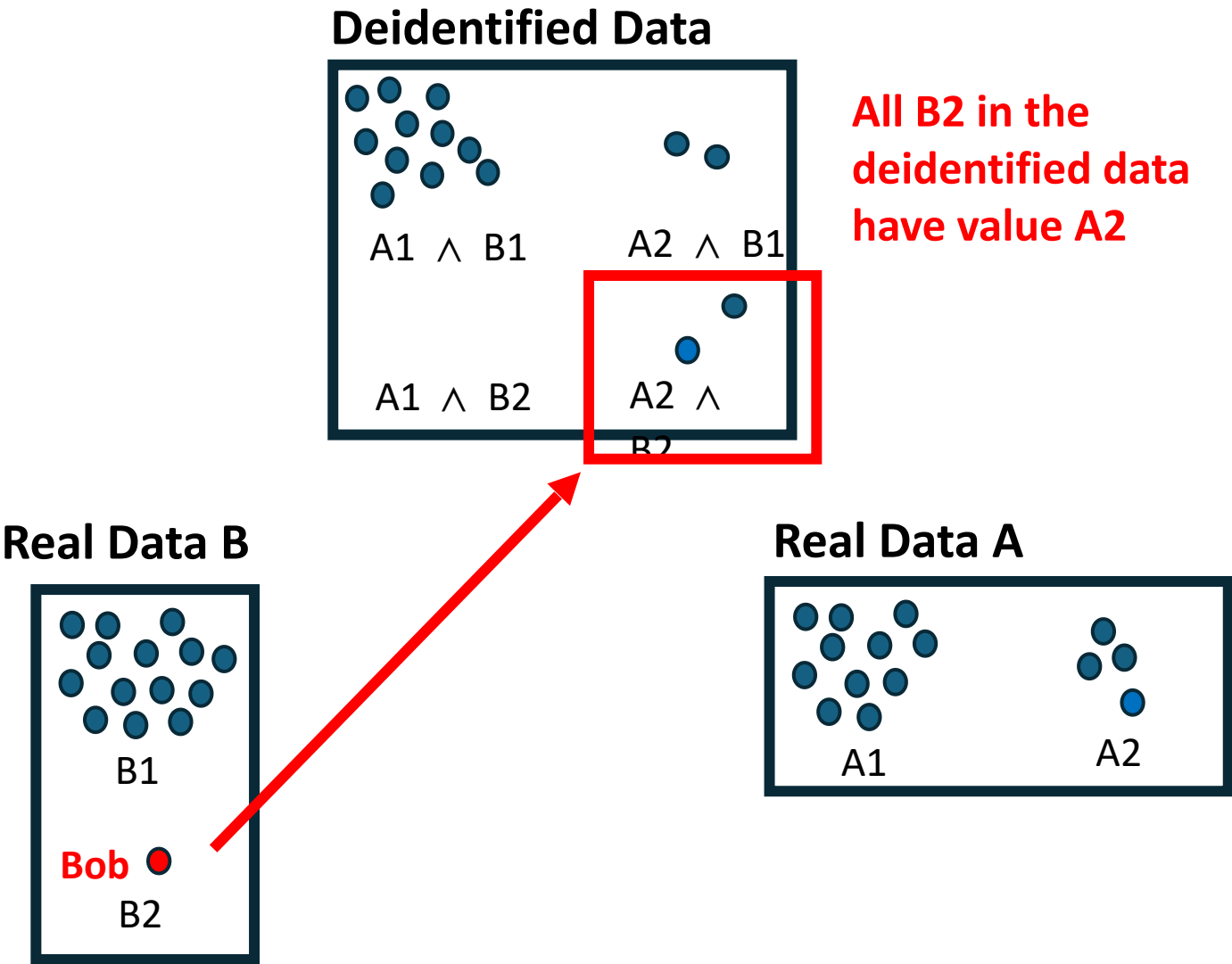
Can I use the released data to.... *find you*?

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[CRC: Unique Exact Match]
- (2) the released deidentified data
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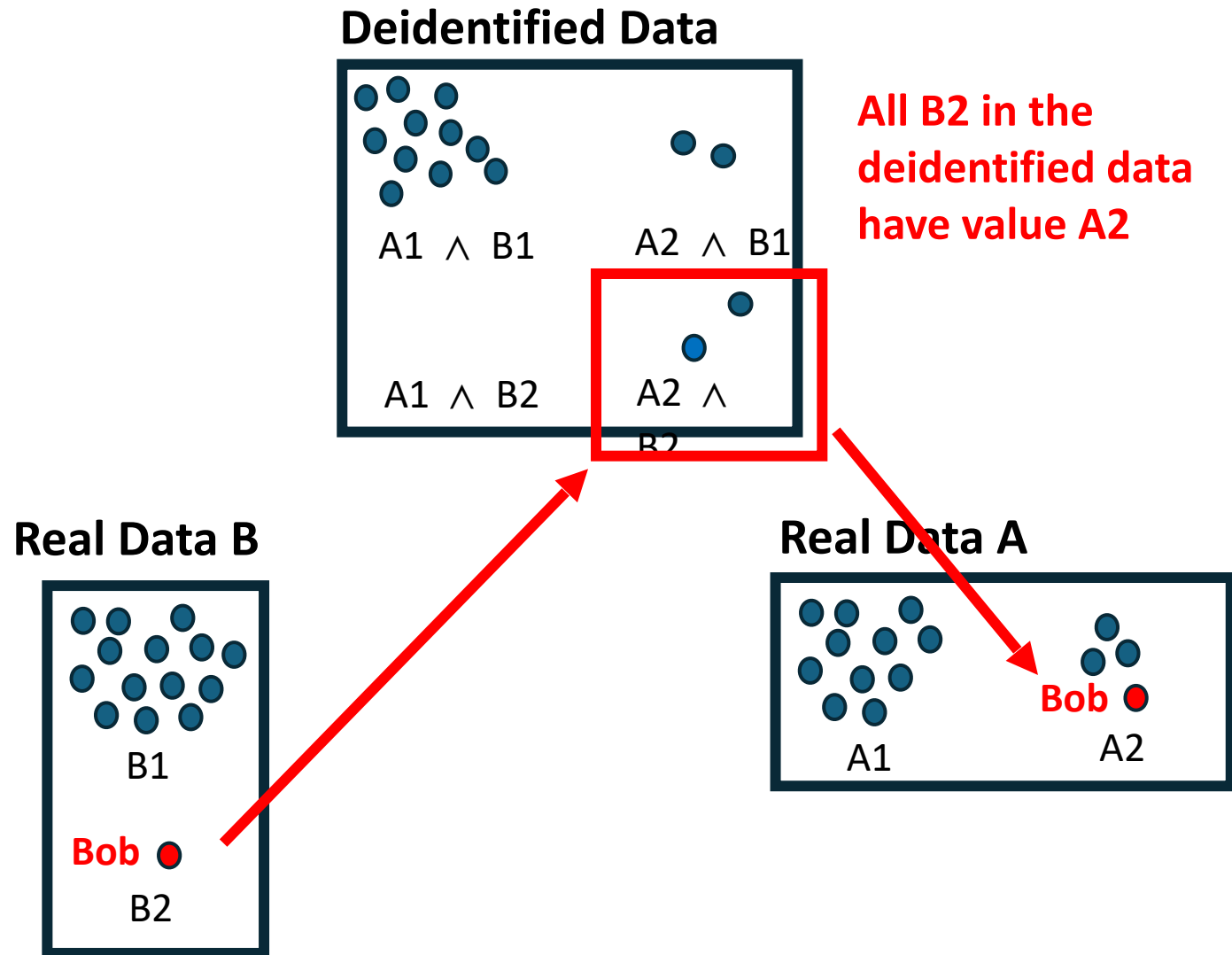
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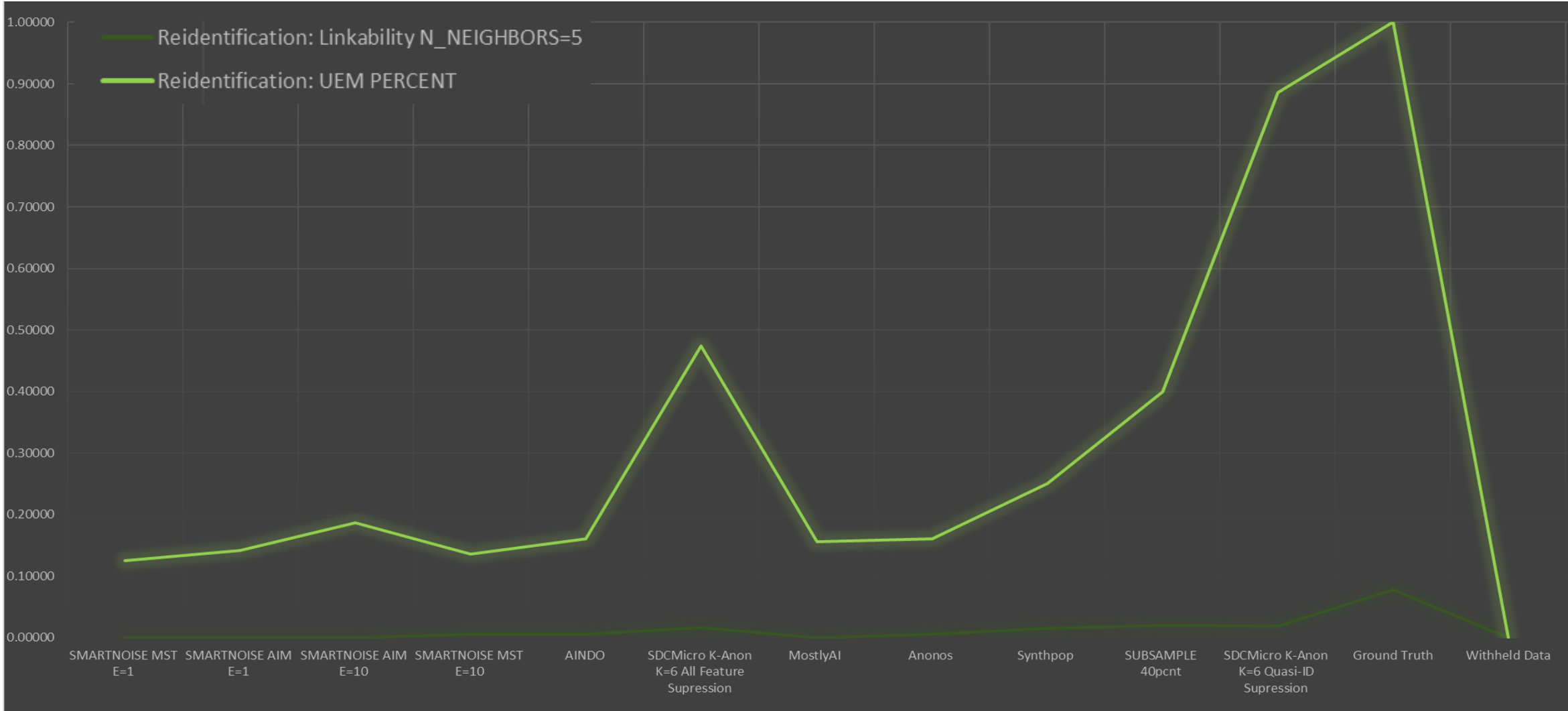
[Anonymizer: Linkability]



Can I use the released data to.... *find you?*

Reidentification Risk: Can the deidentified data help me find your record because...

- (1) **your unique outlier record survived deidentification (as % of unique records)? [Unique Exact Match]**
- (2) **the released data is a bridge between two real data sets with features A and features B? [Linkability]**



Can I use the released data to imagine that I've found you?

Singling Out:

Are there records in the deidentified data that look like they're distinct (unusual) real people? This could happen if the deidentified record is:

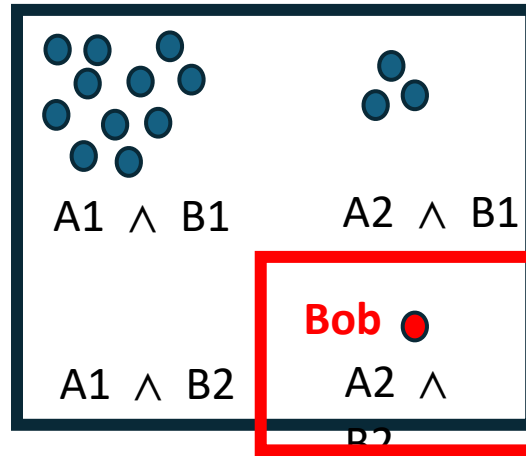
(1) **Closer to a real outlier person**, say Bob, than any other real person is close to Bob.

[Synthcity: Identifiability]

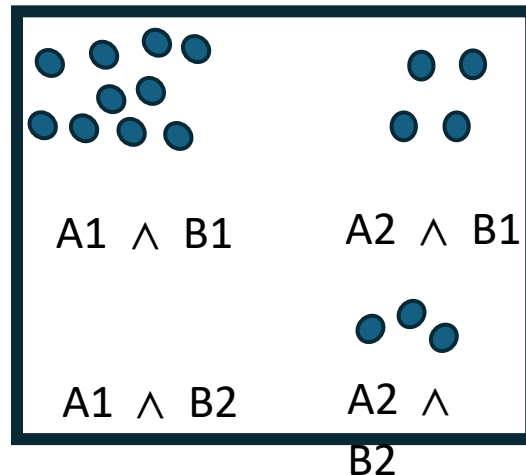
(2) An outlier (unique feature combo) and there's a matching outlier in the real data.

[Anonymeter: Singling Out]

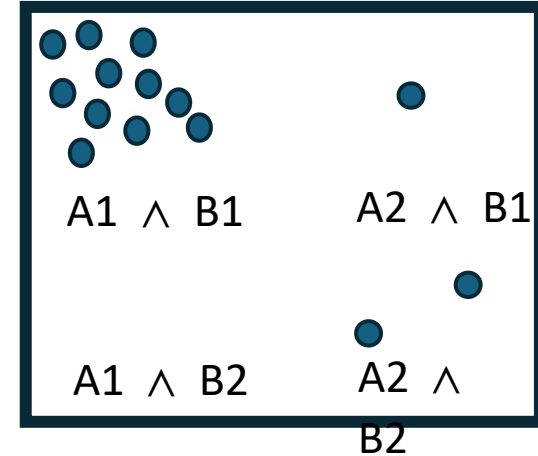
Input Real Data



Withheld Real Data



Deidentified Data



No other real person is like Bob,

Can I use the released data to imagine that I've found you?

Singling Out:

Are there records in the deidentified data that look like they're distinct (unusual) real people? This could happen if the deidentified record is:

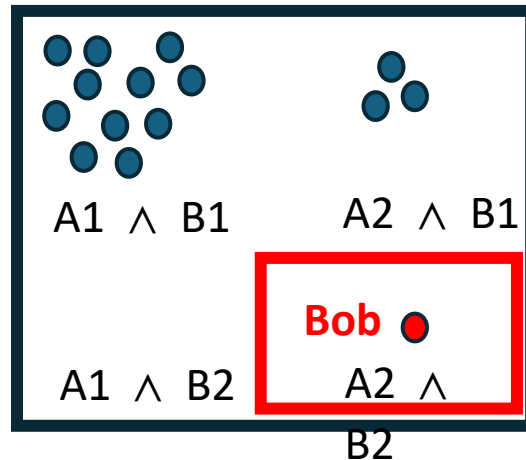
(1) **Closer to a real outlier person**, say Bob, than any other real person is close to Bob.

[Synthcity: Identifiability]

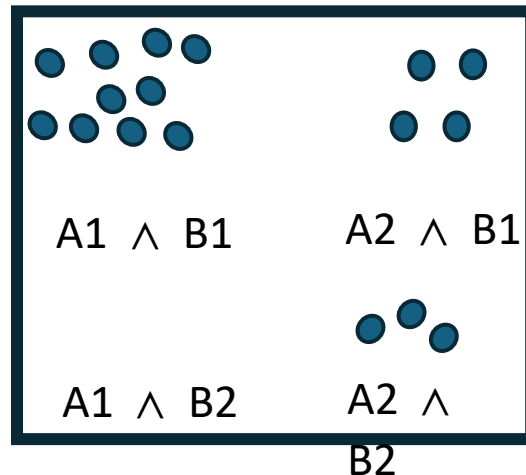
(2) An outlier (unique feature combo) and there's a matching outlier in the real data.

[Anonymizer: Singling Out]

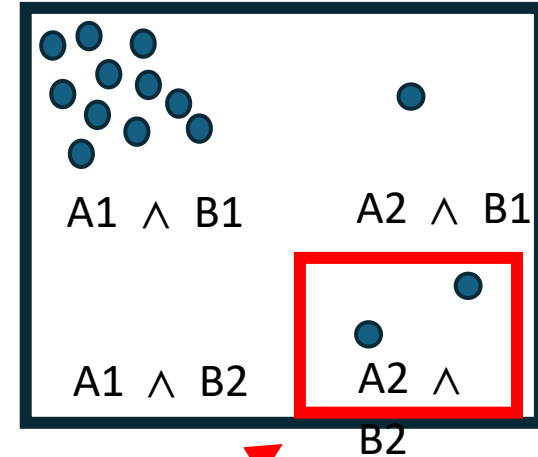
Input Real Data



Withheld Real Data



Deidentified Data



No other real person is like Bob, but there's two synthetic people awfully close to Bob—

They're too close!

Can I use the released data to imagine that I've found you?

Singling Out:

Are there records in the deidentified data that look like they're distinct (unusual) real people? This could happen if the deidentified record is:

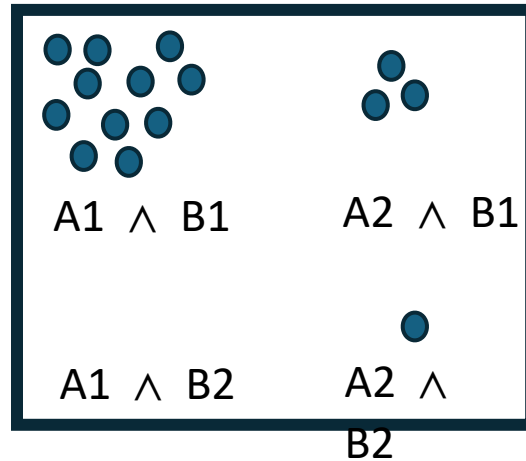
(1) Closer to a real outlier person, say Bob, than any other real person is close to Bob.

[Synthcity: Identifiability]

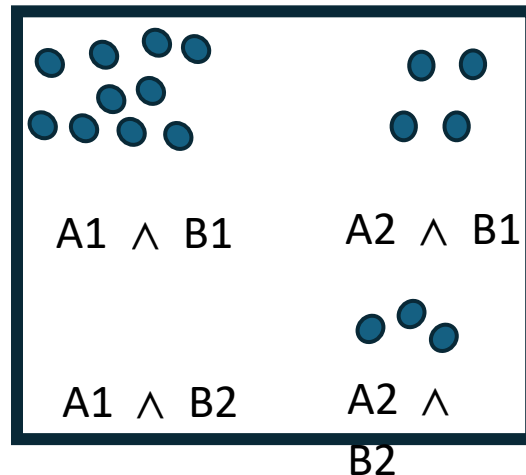
(2) A **synthetic outlier (unique feature combo)** and there's also a matching outlier in the real data.

[Anonymeter: Singling Out]

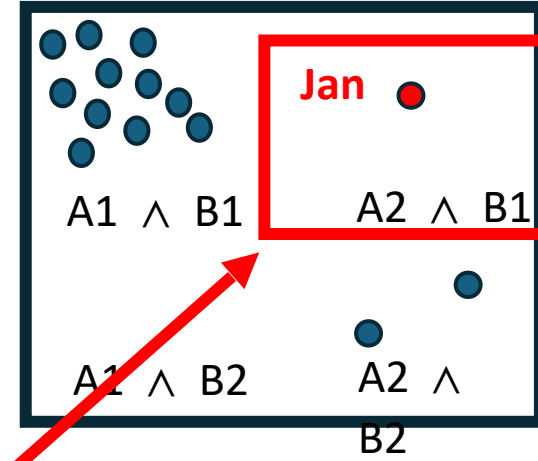
Input Real Data



Withheld Real Data



Deidentified Data



No other fake person is like Jan,

Can I use the released data to imagine that I've found you?

Singling Out:

Are there records in the deidentified data that look like they're distinct (unusual) real people? This could happen if the deidentified record is:

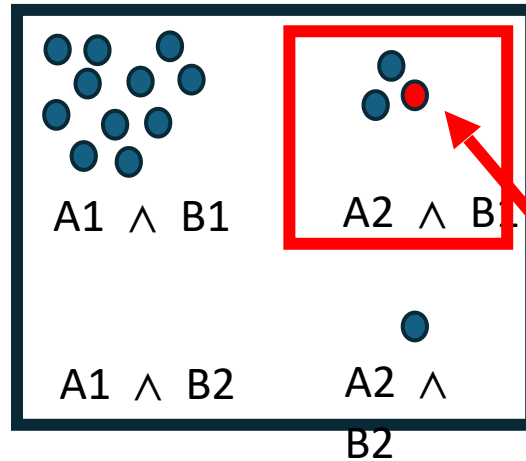
(1) Closer to a real outlier person, say Bob, than any other real person is close to Bob.

[Synthcity: Identifiability]

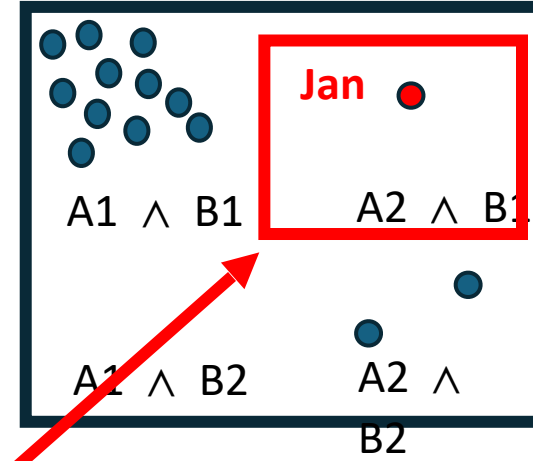
(2) A **synthetic outlier (unique feature combo)** and there's also a matching outlier in the real data.

[Anonymizer: Singling Out]

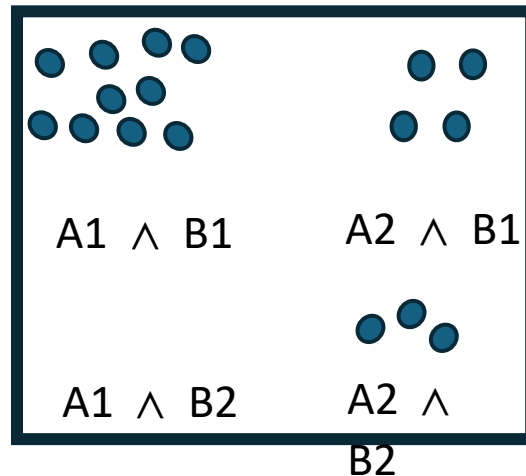
Input Real Data



Deidentified Data



Withheld Real Data



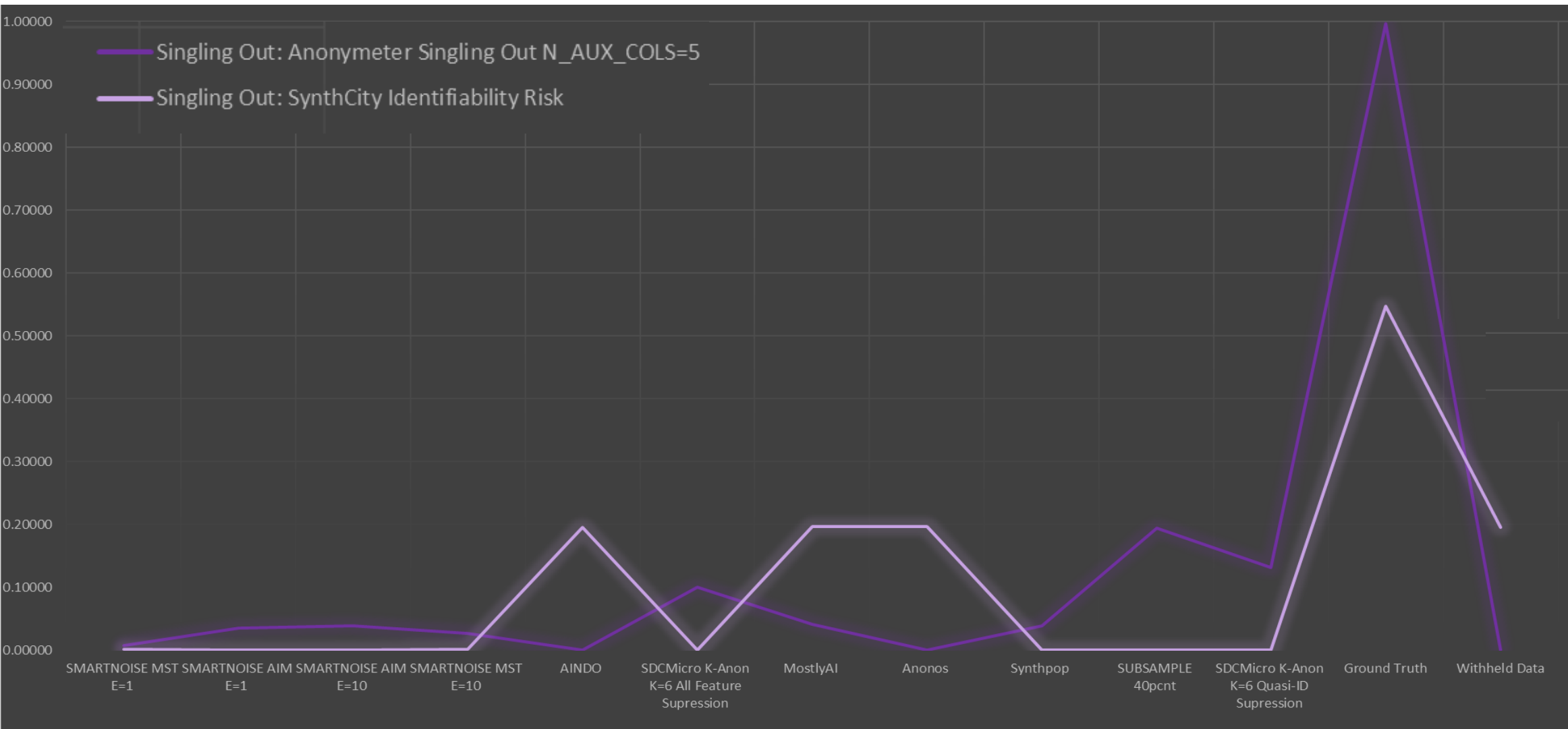
No other fake person is like Jan with A2 and B1.

And there's a real person that matches Jan's unique feature combination.

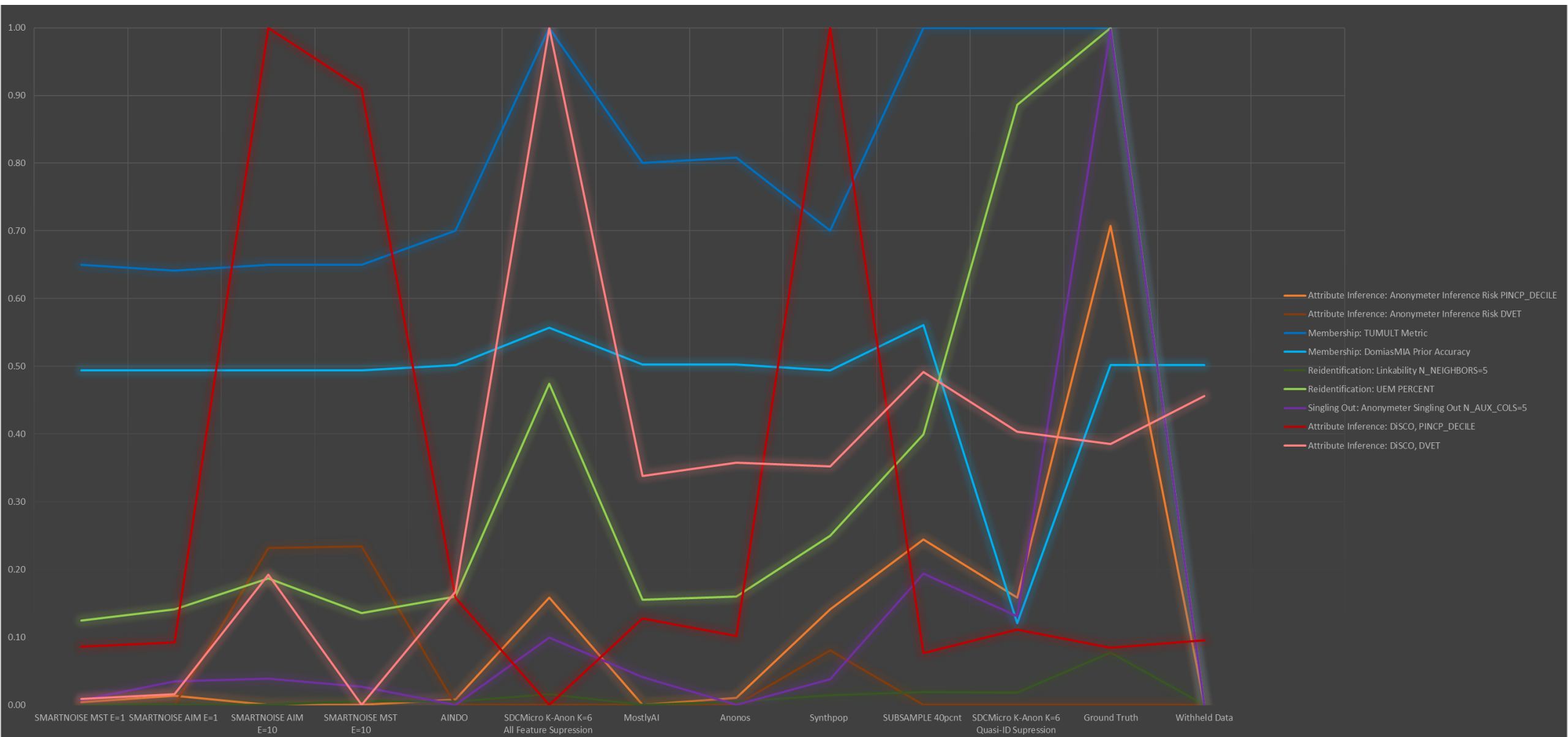
Can I use the released data to imagine that I've found you?

Singling Out: Are there records in the deidentified data that look like they might be distinct real people?

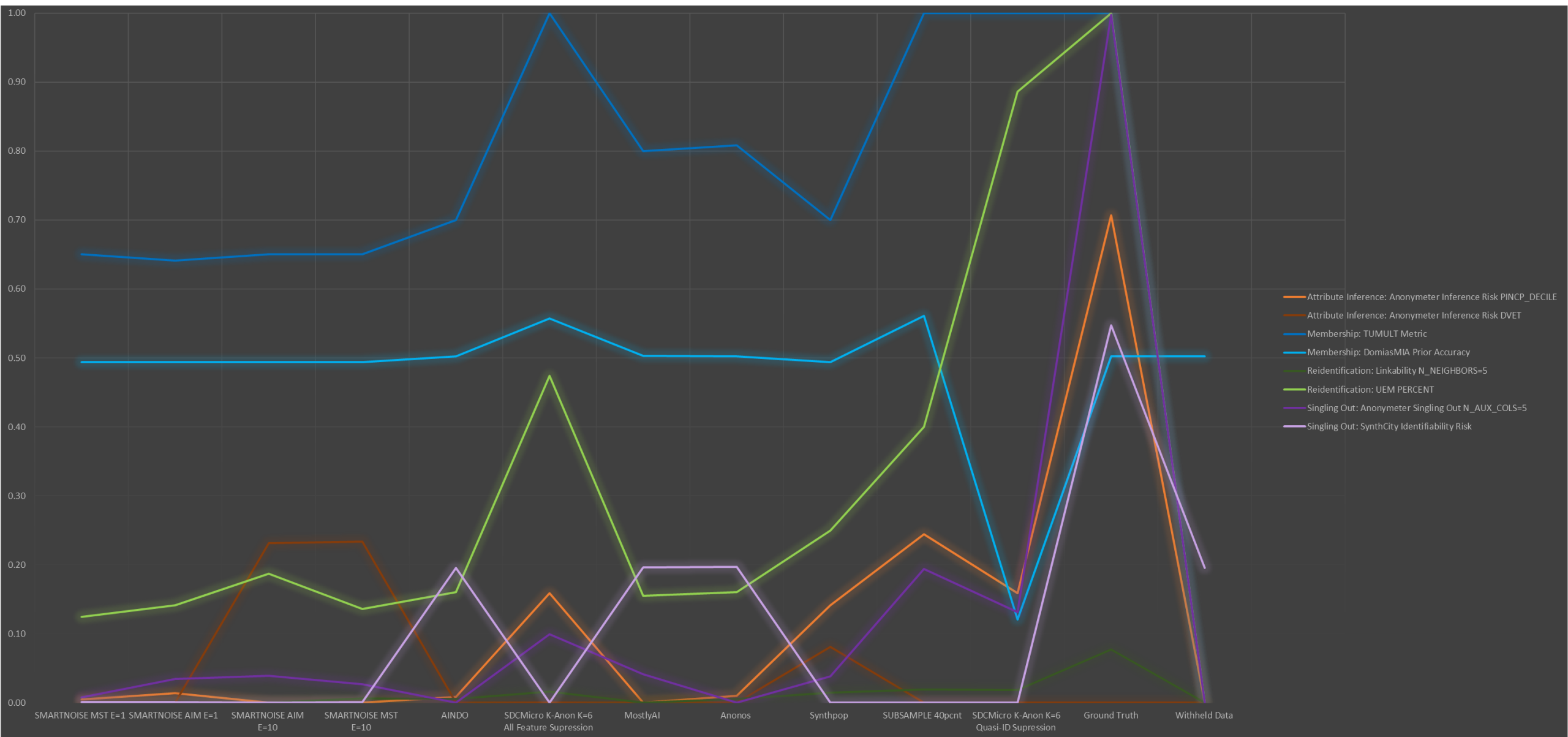
- (1) Fake records closer to a real outlier person than any other real person is. [Synthcity: Identifiability]
- (2) Fake outlier records (unique feature combo) with matching real records [Anonymeter: Singling Out]



So what's private? Is there consensus?



Is there consensus? ...what if we only look at author-tuned metrics?

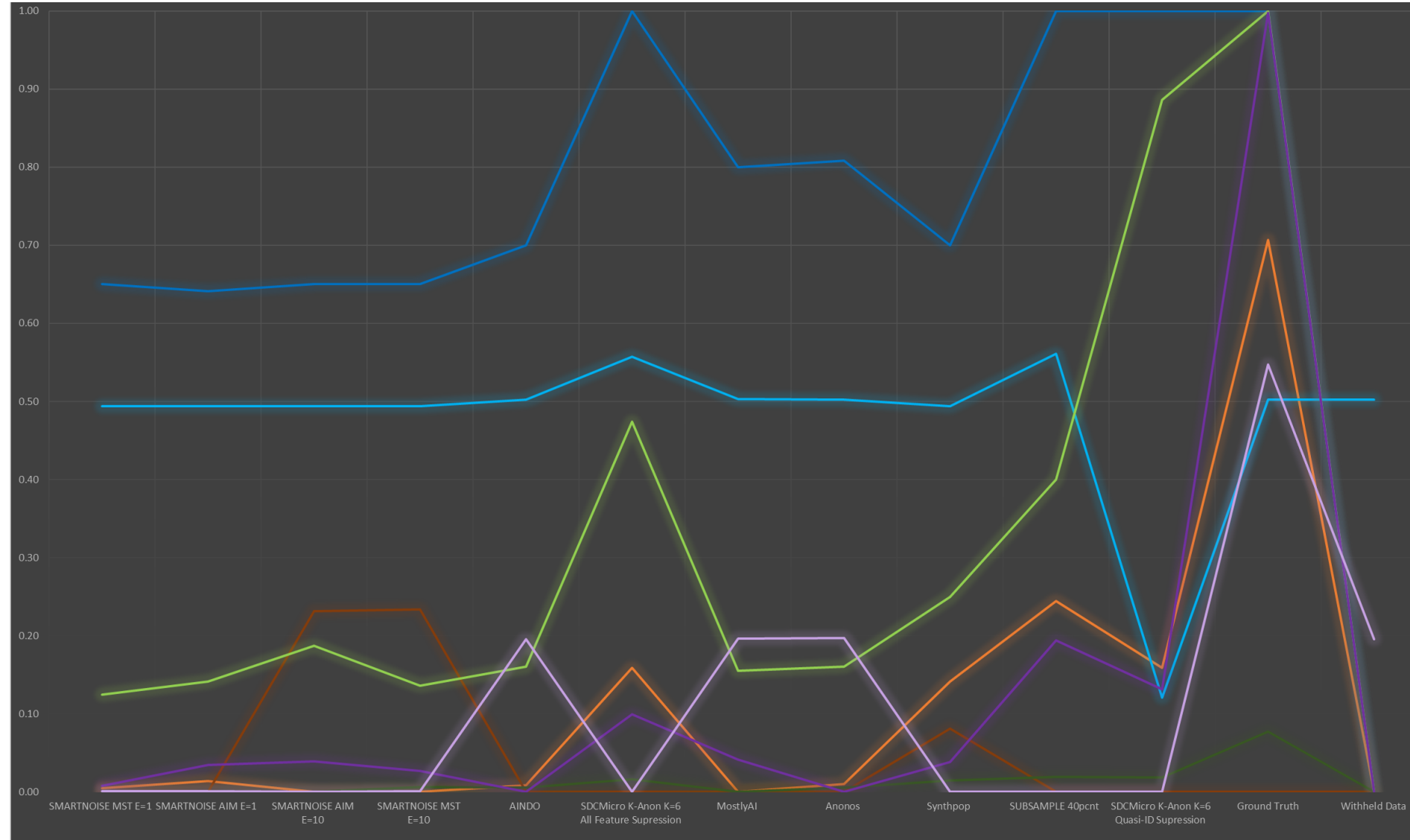


Tuning and configuring privacy metrics (open questions):

Measuring Privacy for Deidentified Data:

Possibly we need to check how we're configuring the metrics:

- Similarity between records with mixed numerical and categorical features
- Quasi-identifiers
- KNN pool size (k)
- etc...



Collaborative Research Cycle: *Red-Teaming* Deidentification Algorithms

Measuring Privacy for Deidentified Data:

Possibly we need to check how we're configuring the metrics.

...or maybe we should just see if any of you can *hack it?*

Next up:

We will be running a community Red Team Exercise in Spring of 2025

Between now and then, we'll be hosting Privacy Metric Discussion Hours, where you can learn the details of the metrics we just flew through, and meet metric designers. **Sign up on our site to keep in touch!**

Thank you!
Questions?

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